



MC3x063A 1.5-A Peak Boost/Buck/Inverting Switching Regulators

1 Features

- Wide Input Voltage Range: 3 V to 40 V
- High Output Switch Current: Up to 1.5 A
- Adjustable Output Voltage
- Oscillator Frequency Up to 100 kHz
- Precision Internal Reference: 2%
- Short-Circuit Current Limiting
- Low Standby Current

2 Applications

- Blood Gas Analyzers: Portable
- Cable Solutions
- HMI's (Human Machine Interfaces)
- Telecommunications
- Portable Devices
- Consumer & Computing
- Test & Measurement

3 Description

The MC33063A and MC34063A devices are easy-to-use ICs containing all the primary circuitry needed for building simple DC-DC converters. These devices primarily consist of an internal temperature-compensated reference, a comparator, an oscillator, a PWM controller with active current limiting, a driver, and a high-current output switch. Thus, the devices require minimal external components to build converters in the boost, buck, and inverting topologies.

The MC33063A device is characterized for operation from -40°C to 85°C , while the MC34063A device is characterized for operation from 0°C to 70°C .

FigureCaption

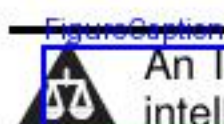
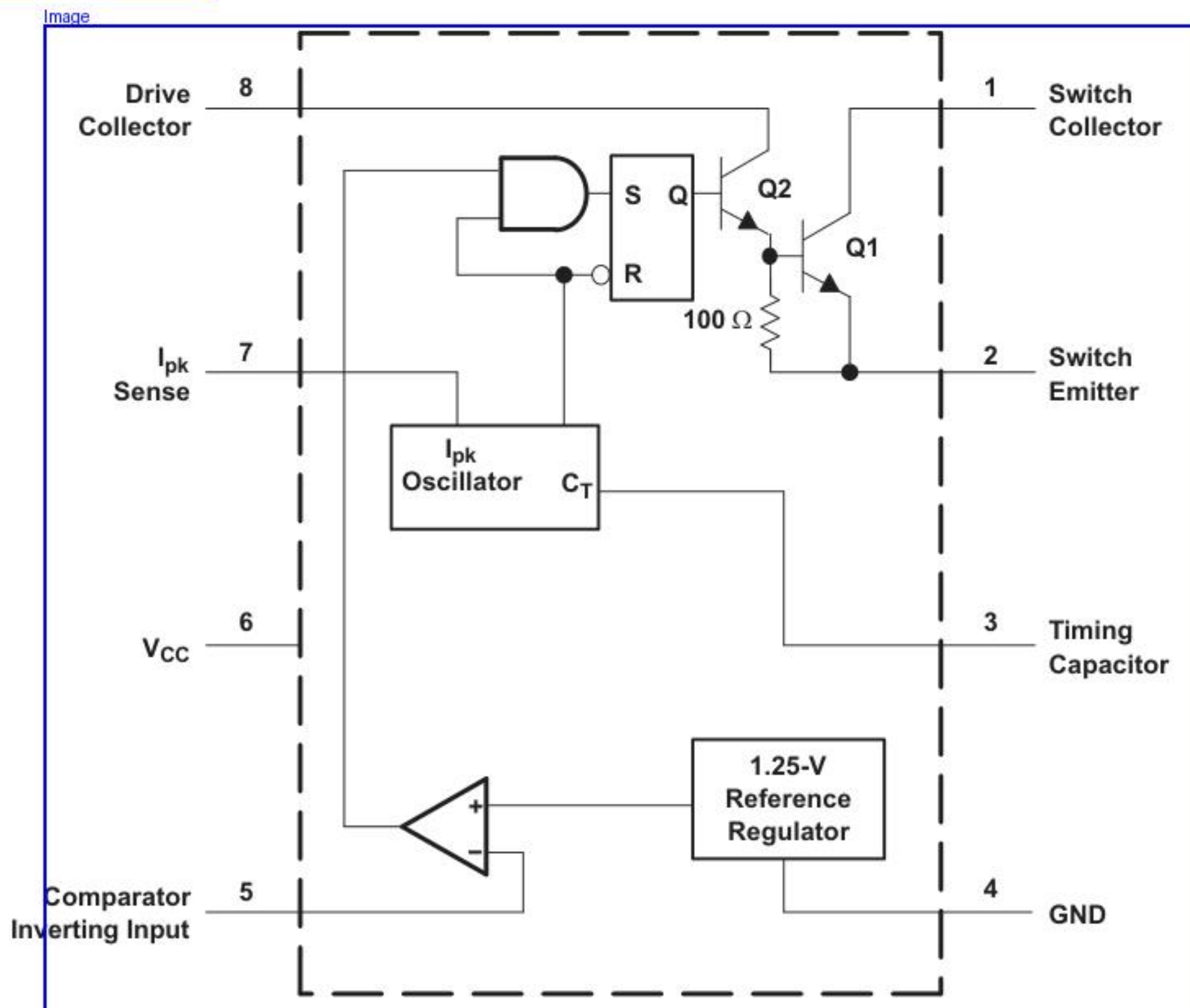
Device Information⁽¹⁾

Device Information		
Part Number	Package (Pin)	Body Size
MC3x063A	SOIC (8)	4.90 mm × 3.91 mm
	SON (8)	4.00 mm × 4.00 mm
	PDIP (8)	9.81 mm × 6.35 mm

List Item

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Title



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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Title

5 Revision History

Title

Changes from Revision M (January 2011) to Revision N

UncategorizedText

Page

UncategorizedText

Added Applications, Device Information table, Pin Functions table, ESD Ratings table, Thermal Information table, Feature Description section, Device Functional Modes, Application and Implementation section, Power Supply Recommendations section, Layout section, Device and Documentation Support section, and Mechanical, Packaging, and Orderable Information section.

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UncategorizedText

Deleted Ordering information table.

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Title

7 Specifications

Title

7.1 Absolute Maximum Ratings

Narrative Text

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

Table

			MIN	MAX	UNIT
V _{CC}	Supply voltage			40	V
V _{IR}	Comparator inverting input voltage range		−0.3	40	V
V _{C(switch)}	Switch collector voltage			40	V
V _{E(switch)}	Switch emitter voltage	V _{PIN1} = 40 V		40	V
V _{CE(switch)}	Switch collector to switch emitter voltage			40	V
V _{C(driver)}	Driver collector voltage			40	V
I _{C(driver)}	Driver collector current			100	mA
I _{SW}	Switch current			1.5	A
T _J	Operating virtual junction temperature			150	°C
T _{stg}	Storage temperature range		−65	150	°C

List Item

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

Title

7.2 ESD Ratings

Table

		VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins ⁽¹⁾	2500
		Charged device model (CDM), per JEDEC specification JESD22-C101, all pins ⁽²⁾	1500

List Item

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

Title

7.3 Recommended Operating Conditions

Table

		MIN	MAX	UNIT
V _{CC}	Supply voltage	3	40	V
T _A	Operating free-air temperature	MC33063A	−40	85
		MC34063A	0	70

Title

7.4 Thermal Information

Table

	THERMAL METRIC ⁽¹⁾	MC33063A			UNIT
		D	DRJ	P	
		8 PINS			
R _{θJA}	Junction-to-ambient thermal resistance	97	41	85	°C/W

Narrative Text

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report (SPRA953).

Title

7.5 Electrical Characteristics—Oscillator

Narrative Text

V_{CC} = 5 V, I_A = full operating range (unless otherwise noted) (see block diagram)

	PARAMETER	TEST CONDITIONS	T _A	MIN	TYP	MAX	UNIT
f _{osc}	Oscillator frequency	V _{PIN5} = 0 V, C _T = 1 nF	25°C	24	33	42	kHz
I _{chg}	Charge current	V _{CC} = 5 V to 40 V	25°C	24	35	42	μA
I _{dischg}	Discharge current	V _{CC} = 5 V to 40 V	25°C	140	220	260	μA
I _{dischg} /I _{chg}	Discharge-to-charge current ratio	V _{PIN7} = V _{CC}	25°C	5.2	6.5	7.5	—
V _{Ipk}	Current-limit sense voltage	I _{dischg} = I _{chg}	25°C	250	300	350	mV

7.6 Electrical Characteristics—Output Switch

Narrative Text
 $V_{CE} = 5\text{ V}$, $I_A =$ full operating range (unless otherwise noted) (see block diagram) ⁽¹⁾

PARAMETER	TEST CONDITIONS	T_A	MIN	TYP	MAX	UNIT
$V_{CE(sat)}$	Saturation voltage – Darlington connection	$I_{SW} = 1\text{ A}$, pins 1 and 8 connected	Full range	1	1.3	V
$V_{CE(sat)}$	Saturation voltage – non-Darlington connection ⁽²⁾	$I_{SW} = 1\text{ A}$, $R_{PIN8} = 82\ \Omega$ to V_{CC} , forced $\beta \sim 20$	Full range	0.45	0.7	V
h_{FE}	DC current gain	$I_{SW} = 1\text{ A}$, $V_{CE} = 5\text{ V}$	25°C	50	75	—
$I_{C(off)}$	Collector off-state current	$V_{CE} = 40\text{ V}$	Full range	0.01	100	μA

List Item
(1) Low duty-cycle pulse testing is used to maintain junction temperature as close to ambient temperature as possible.

(2) In the non-Darlington configuration, if the output switch is driven into hard saturation at low switch currents ($\leq 300\text{ mA}$) and high driver currents ($\geq 30\text{ mA}$), it may take up to $2\ \mu\text{s}$ for the switch to come out of saturation. This condition effectively shortens the off time at frequencies $\geq 30\text{ kHz}$, becoming magnified as temperature increases. The following output drive condition is recommended in the non-Darlington configuration:

Narrative Text
Forced β of output switch = $I_{C,SW} / (I_{C,driver} - \sim\text{mA}) \geq 10$, where $\sim\text{mA}$ is required by the $100\text{-}\Omega$ resistor in the emitter of the driver to forward bias the V_{be} of the switch.

7.7 Electrical Characteristics—Comparator

Narrative Text
 $V_{CE} = 5\text{ V}$, $T_A =$ full operating range (unless otherwise noted) (see block diagram)

PARAMETER	TEST CONDITIONS	T_A	MIN	TYP	MAX	UNIT
V_{th}	Threshold voltage	25°C	1.225	1.25	1.275	V
		Full range	1.21		1.29	
ΔV_{th}	Threshold-voltage line regulation	$V_{CC} = 5\text{ V to } 40\text{ V}$	Full range	1.4	5	mV
I_{IB}	Input bias current	$V_{IN} = 0\text{ V}$	Full range	–20	–400	nA

7.8 Electrical Characteristics—Total Device

Narrative Text
 $V_{CC} = 5\text{ V}$, $T_A =$ full operating range (unless otherwise noted) (see block diagram)

PARAMETER	TEST CONDITIONS	T_A	MIN	MAX	UNIT
I_{CC}	Supply current $V_{CC} = 5\text{ V to } 40\text{ V}$, $C_T = 1\text{ nF}$, $V_{PIN7} = V_{CC}$, $V_{PIN5} > V_{th}$, $V_{PIN2} = \text{GND}$, All other pins open	Full range		4	mA

Title
7.9 Typical Characteristics

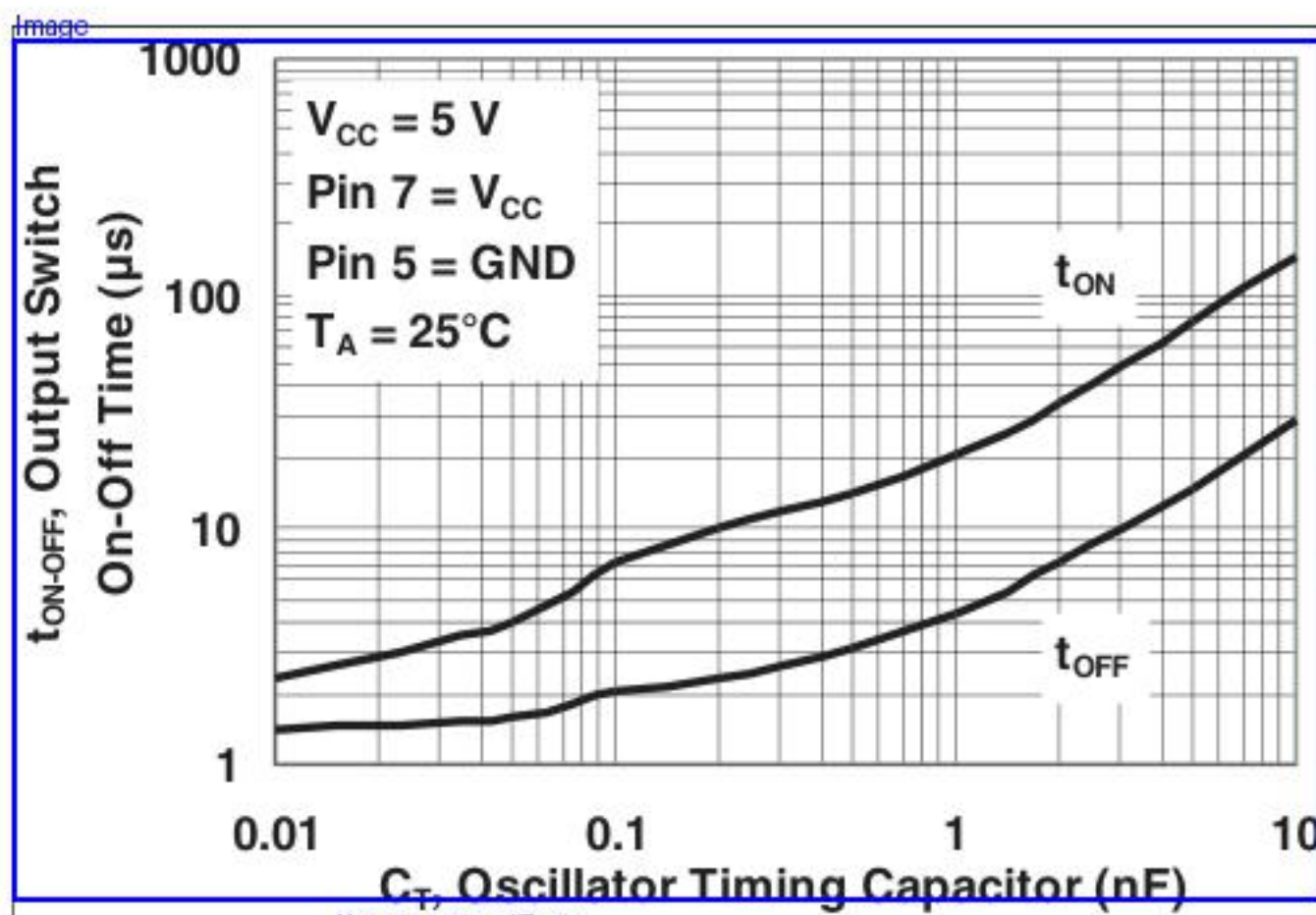


Figure 1. Output Switch On-Off Time vs Oscillator Timing Capacitor

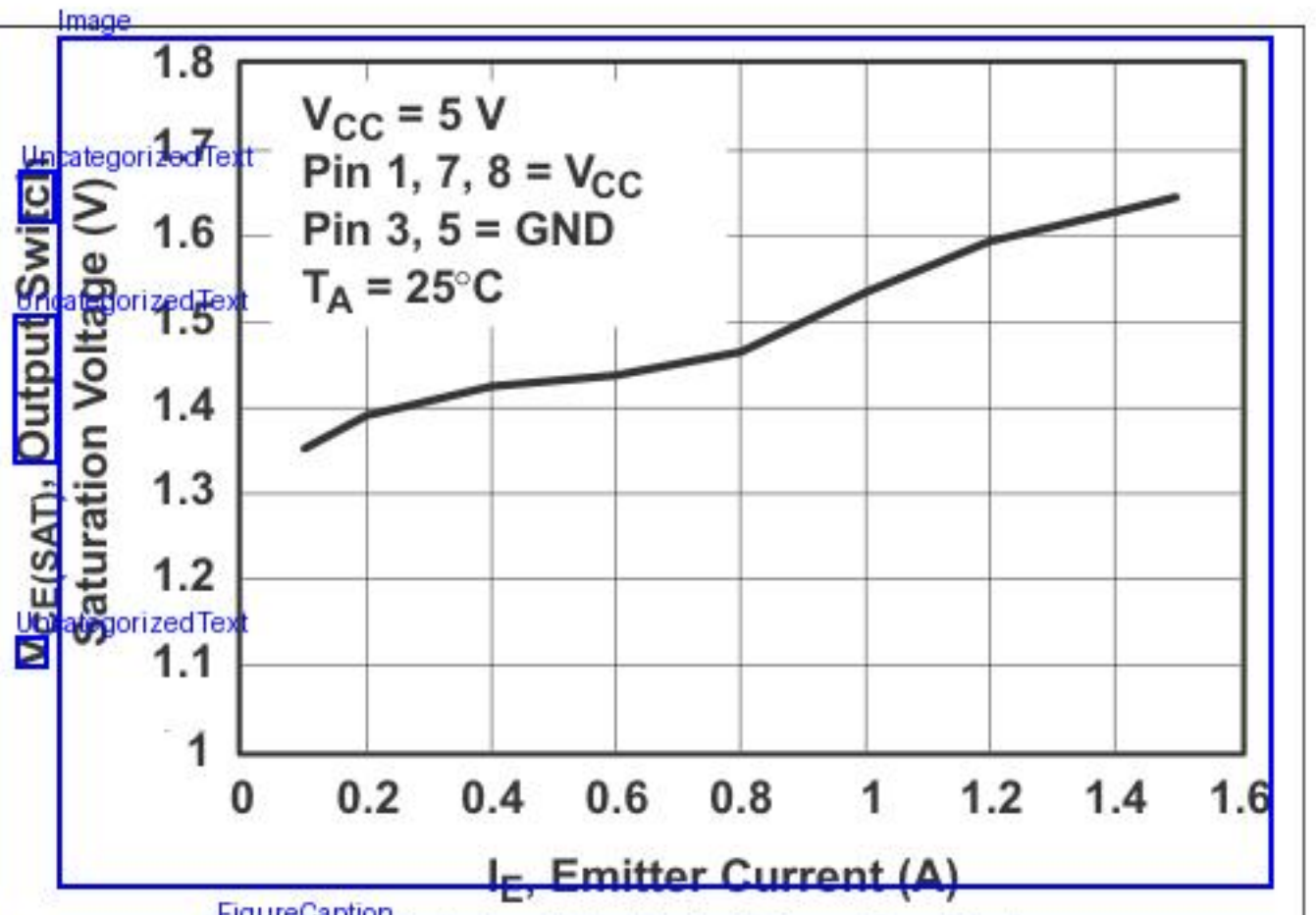


Figure 2. Output Switch Saturation Voltage vs Emitter Current (Emitter-Follower Configuration)

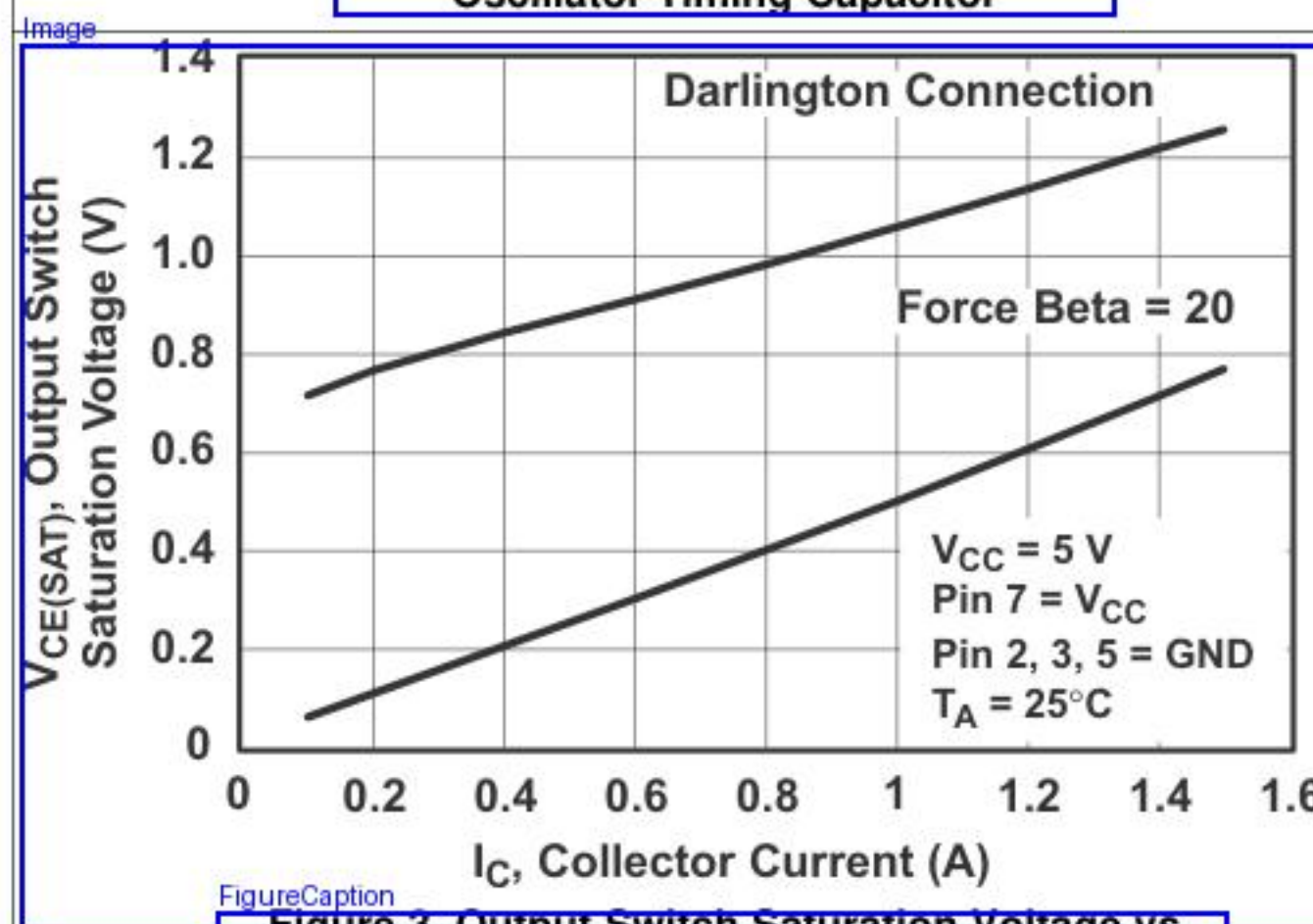


Figure 3. Output Switch Saturation Voltage vs Collector Current (Common-Emitter Configuration)

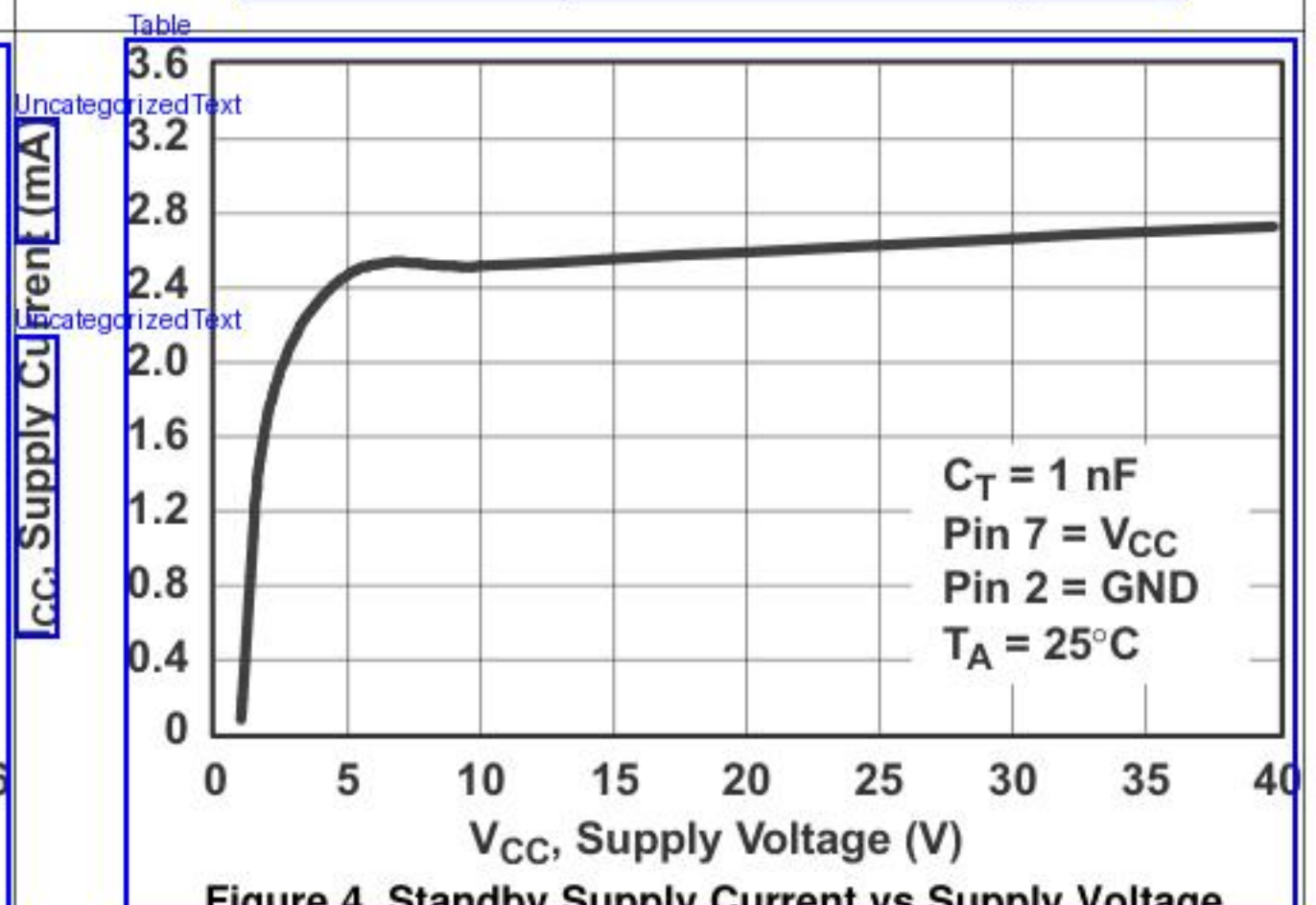


Figure 4. Standby Supply Current vs Supply Voltage

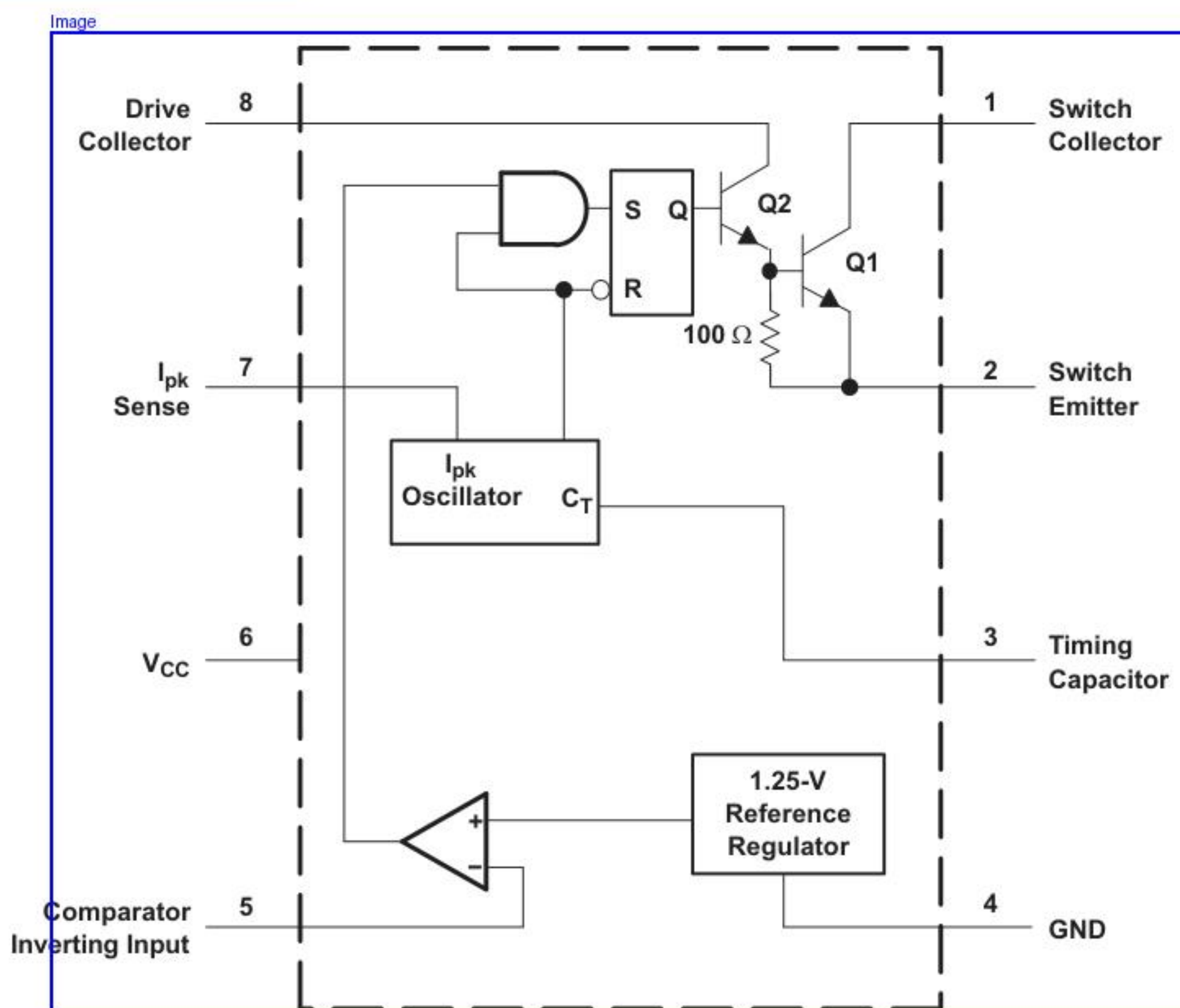
8 Detailed Description

8.1 Overview

The MC33063A and MC34063A devices are easy-to-use ICs containing all the primary circuitry needed for building simple DC-DC converters. These devices primarily consist of an internal temperature-compensated reference, a comparator, an oscillator, a PWM controller with active current limiting, a driver, and a high-current output switch. Thus, the devices require minimal external components to build converters in the boost, buck, and inverting topologies.

The MC33063A device is characterized for operation from -40°C to 85°C , while the MC34063A device is characterized for operation from 0°C to 70°C .

8.2 Functional Block Diagram



8.3 Feature Description

- Wide Input Voltage Range: 3 V to 40 V
- High Output Switch Current: Up to 1.5 A
- Adjustable Output Voltage
- Oscillator Frequency Up to 100 kHz
- Precision Internal Reference: 2%
- Short-Circuit Current Limiting
- Low Standby Current

8.4 Device Functional Modes

8.4.1 Standard operation

Based on the application, the device can be configured in multiple different topologies. See the [Application and Implementation](#) section for how to configure the device in several different operating modes.

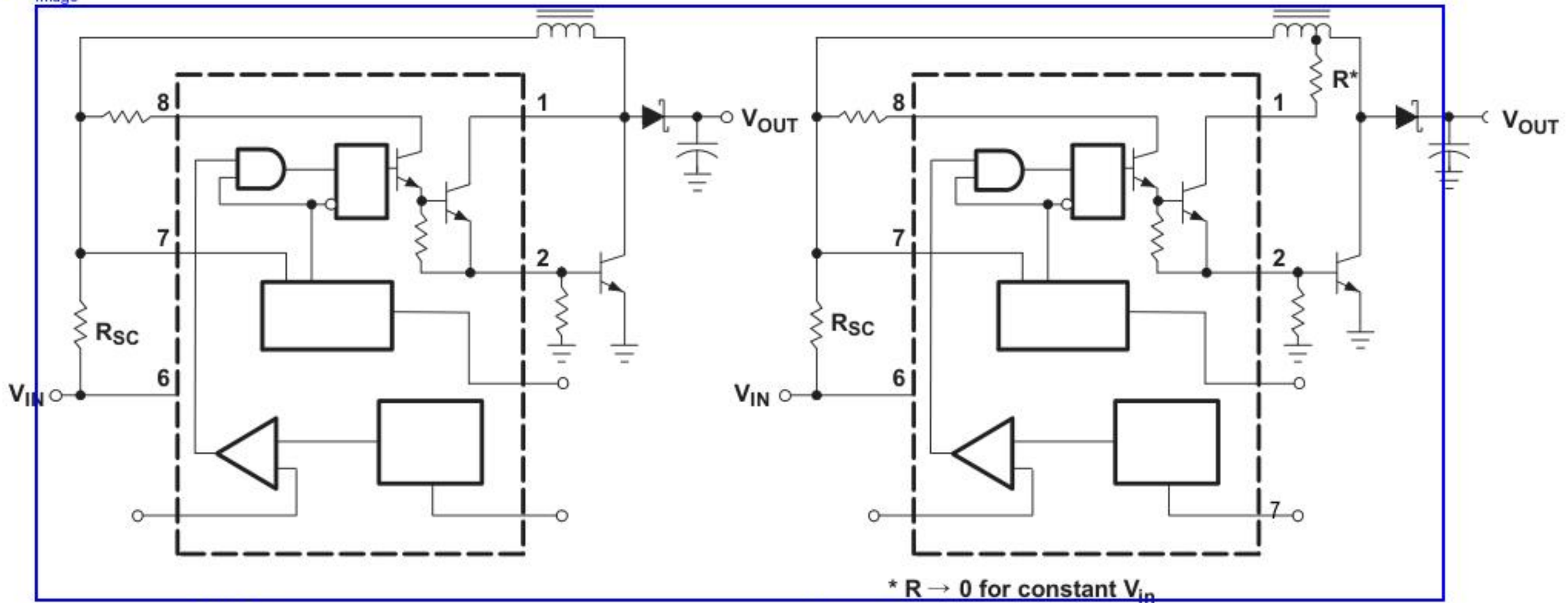
9 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

9.1.1 External Switch Configurations for Higher Peak Current



a) EXTERNAL npn SWITCH

b) EXTERNAL npn SATURATED SWITCH (see Note A)

- A. If the output switch is driven into hard saturation (non-Darlington configuration) at low switch currents (≤ 300 mA) and high driver currents (≥ 30 mA), it may take up to 2 μ s to come out of saturation. This condition will shorten the off time at frequencies ≥ 30 kHz and is magnified at high temperatures. This condition does not occur with a Darlington configuration because the output switch cannot saturate. If a non-Darlington configuration is used, the output drive configuration in Figure 7b is recommended.

Figure 5. Boost Regulator Connections for I_C Peak Greater Than 1.5 A

Title
Typical Application (continued)

Title
9.2.1.1 Design Requirements

Narrative Text
The user must determine the following desired parameters:

Narrative Text
 V_{sat} = Saturation voltage of the output switch

Narrative Text
 V_F = Forward voltage drop of the chosen output rectifier

ListItem
The following power-supply parameters are set by the user:

Narrative Text
 V_{in} = Nominal input voltage

Narrative Text
 V_{out} = Desired output voltage

Narrative Text
 I_{out} = Desired output current

Narrative Text
 f_{min} = Minimum desired output switching frequency at the selected values of V_{in} and I_{out}

Narrative Text
 V_{ripple} = Desired peak-to-peak output ripple voltage. The ripple voltage directly affects the line and load regulation and, thus, must be considered. In practice, the actual capacitor value should be larger than the calculated value, to account for the capacitor's equivalent series resistance and board layout.

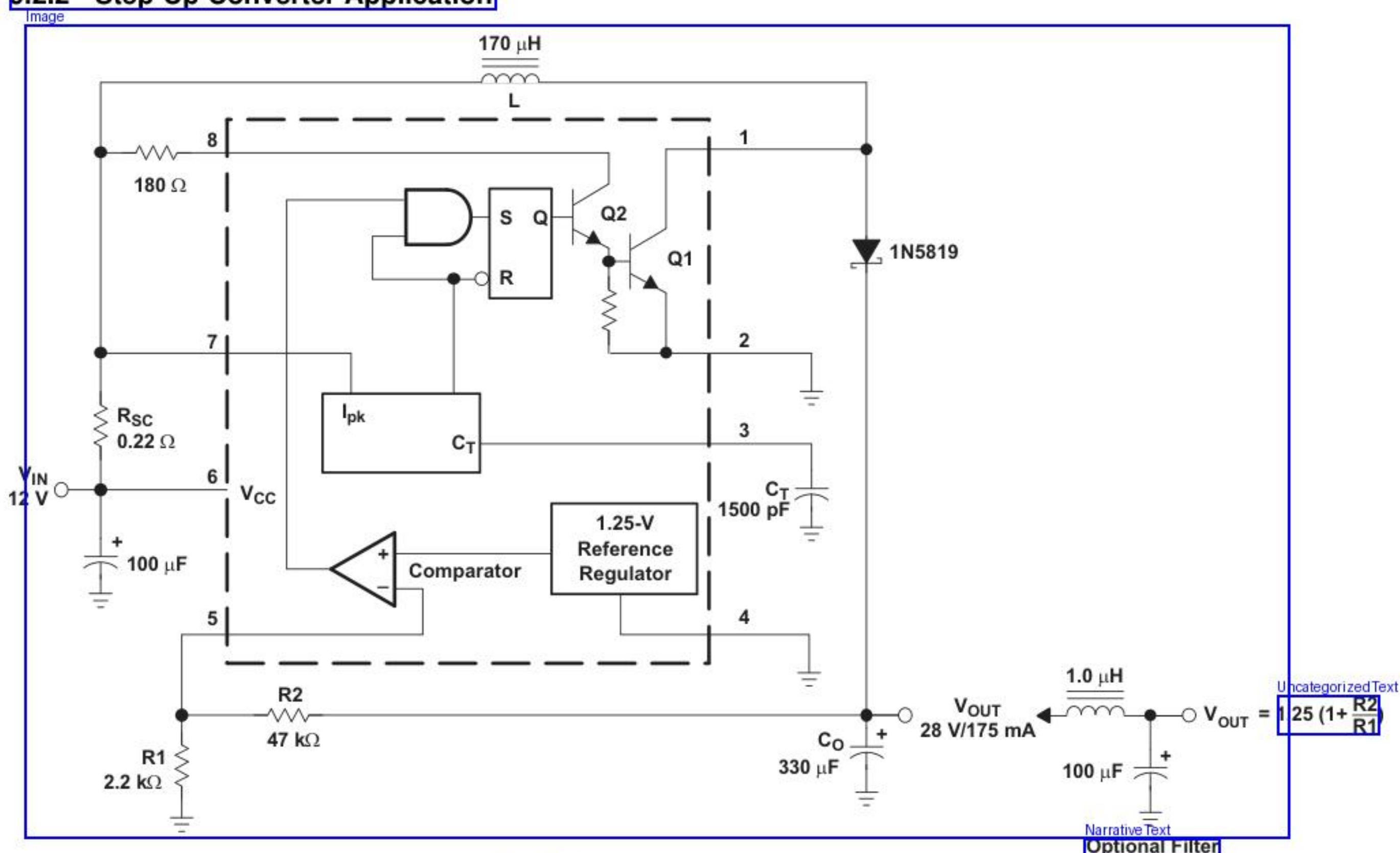
Title
9.2.1.2 Detailed Design Procedure

Table

CALCULATION	VOLTAGE INVERTING
t_{on}/t_{off}	$\frac{ V_{out} + V_F}{V_{in} - V_{sat}}$
$(t_{on} + t_{off})$	$\frac{1}{f}$
t_{off}	$\frac{t_{on} + t_{off}}{\frac{t_{on}}{t_{off}} + 1}$
t_{on}	$(t_{on} + t_{off}) - t_{off}$
C_T	$4 \times 10^{-5} t_{on}$
$I_{pk(switch)}$	$2I_{out(max)} \left(\frac{t_{on}}{t_{off}} + 1 \right)$
R_{SC}	$\frac{0.3}{I_{pk(switch)}}$
$L_{(min)}$	$\left(\frac{(V_{in(min)} - V_{sat})}{I_{pk(switch)}} \right) t_{on(max)}$
C_O	$9 \frac{I_{out} t_{on}}{V_{ripple(pp)}}$
V_{out}	$-1.25 \left(1 + \frac{R2}{R1} \right)$

See Figure 8

Title
9.2.2 Step-Up Converter Application



Title
9.2.2.1 Design Requirements

NarrativeText
 The user must determine the following desired parameters:

NarrativeText
 V_{sat} = Saturation voltage of the output switch

NarrativeText
 V_F = Forward voltage drop of the chosen output rectifier

NarrativeText
 The following power-supply parameters are set by the user:

NarrativeText
 V_{in} = Nominal input voltage

NarrativeText
 V_{out} = Desired output voltage

NarrativeText
 I_{out} = Desired output current

NarrativeText
 f_{min} = Minimum desired output switching frequency at the selected values of V_{in} and I_{out}

NarrativeText
 V_{ripple} = Desired peak-to-peak output ripple voltage. The ripple voltage directly affects the line and load regulation and, thus, must be considered. In practice, the actual capacitor value should be larger than the calculated value, to account for the capacitor's equivalent series resistance and board layout.

Header

MC33063A, MC34063A

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9.2.2.2 Detailed Design Procedure

Table

CALCULATION	STEP UP
t_{on}/t_{off}	$\frac{V_{out} + V_{F-Vin(min)}}{V_{in(min)} - V_{sat}}$
$(t_{on} + t_{off})$	$\frac{1}{f}$
t_{off}	$\frac{t_{on} + t_{off}}{\frac{t_{on}}{t_{off}} + 1}$
t_{on}	$(t_{on} + t_{off}) - t_{off}$
C_T	$4 \times 10^{-5} t_{on}$
$I_{pk(switch)}$	$2I_{out(max)} \left(\frac{t_{on}}{t_{off}} + 1 \right)$
R_{sc}	$\frac{0.3}{I_{pk(switch)}}$
$L_{(min)}$	$\left(\frac{(V_{in(min)} - V_{sat})}{I_{pk(switch)}} \right) t_{on(max)}$
C_O	$9 \frac{I_{out} t_{on}}{V_{ripple(pp)}}$
V_{out}	$1.25 \left(1 + \frac{R2}{R1} \right)$ See Figure 10

Title

9.2.2.3 Application Performance

Table

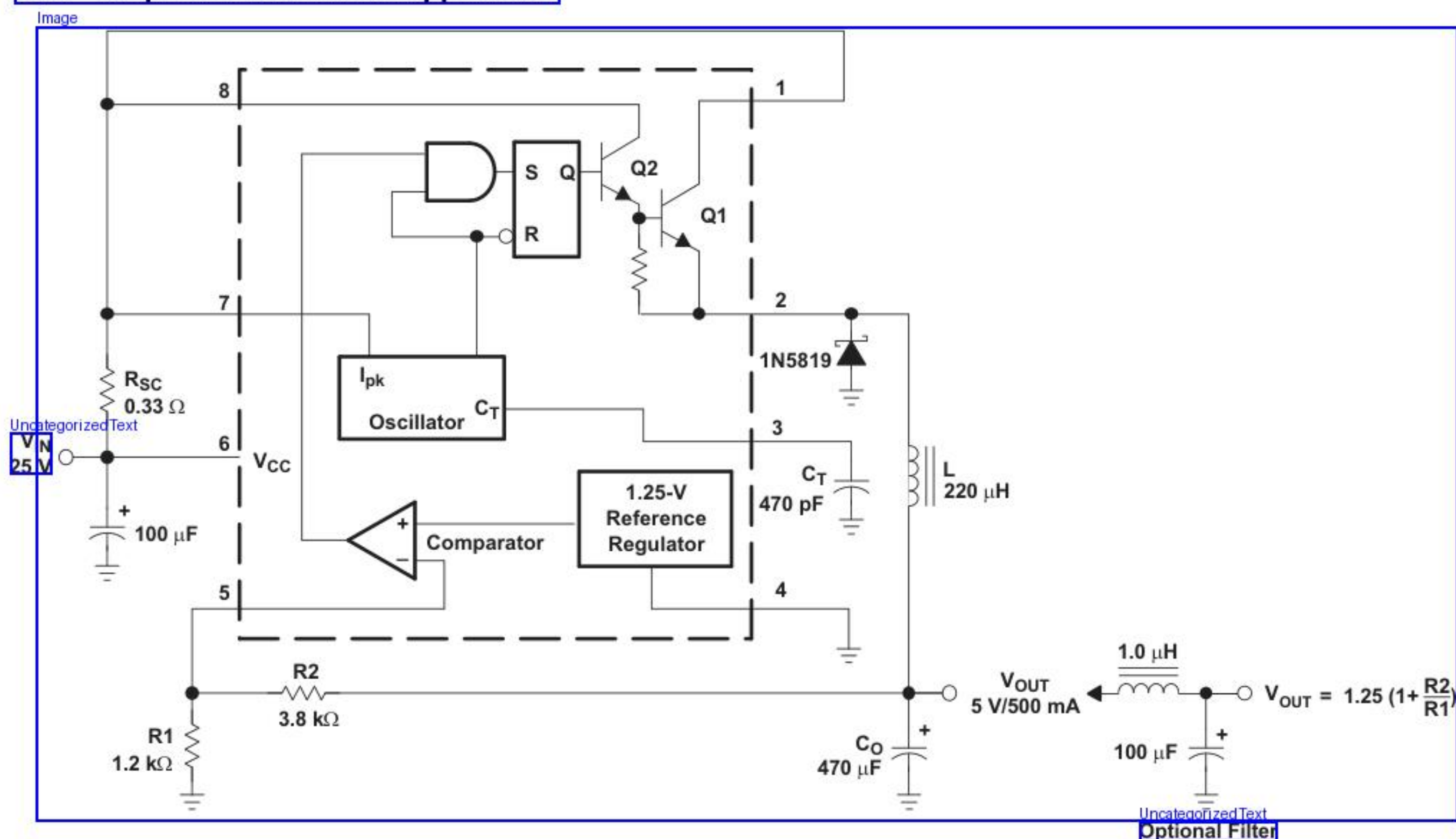
TEST	CONDITIONS	RESULTS
Line regulation	$V_{IN} = 8 \text{ V to } 16 \text{ V}, I_O = 175 \text{ mA}$	$30 \text{ mV} \pm 0.05\%$
Load regulation	$V_{IN} = 12 \text{ V}, I_O = 75 \text{ mA to } 175 \text{ mA}$	$10 \text{ mV} \pm 0.017\%$
Output ripple	$V_{IN} = 12 \text{ V}, I_O = 175 \text{ mA}$	400 mV_{PP}
Efficiency	$V_{IN} = 12 \text{ V}, I_O = 175 \text{ mA}$	87.7%
Output ripple with optional filter	$V_{IN} = 12 \text{ V}, I_O = 175 \text{ mA}$	40 mV_{PP}

Page

Narrative Text

Narrative Text

9.2.3 Step-Down Converter Application



FigureCaption
Figure 11. Step-Down Converter

9.2.3.1 Design Requirements

The user must determine the following desired parameters:

V_{sat} = Saturation voltage of the output switch

V_F = Forward voltage drop of the chosen output rectifier

The following power-supply parameters are set by the user:

V_{in} = Nominal input voltage

V_{out} = Desired output voltage

I_{out} = Desired output current

f_{min} = Minimum desired output switching frequency at the selected values of V_{in} and I_{out}

V_{ripple} = Desired peak-to-peak output ripple voltage. The ripple voltage directly affects the line and load regulation and, thus, must be considered. In practice, the actual capacitor value should be larger than the calculated value, to account for the capacitor's equivalent series resistance and board layout.

Header

MC33063A, MC34063A

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Title

9.2.3.2 Detailed Design Procedure

Table

CALCULATION	STEP DOWN
t_{on}/t_{off}	$\frac{V_{out} + V_F}{V_{in(min)} - V_{sat} - V_{out}}$
$(t_{on} + t_{off})$	$\frac{1}{f}$
t_{off}	$\frac{t_{on} + t_{off}}{\frac{t_{on}}{t_{off}} + 1}$
t_{on}	$(t_{on} + t_{off}) - t_{off}$
C_T	$4 \times 10^{-5} t_{on}$
$I_{pk(switch)}$	$2I_{out(max)}$
R_{SC}	$\frac{0.3}{I_{pk(switch)}}$
$L_{(min)}$	$\left(\frac{(V_{in(min)} - V_{sat} - V_{out})}{I_{pk(switch)}} \right) t_{on(max)}$
C_O	$\frac{I_{pk(switch)} (t_{on} + t_{off})}{8V_{ripple(pp)}}$
V_{out}	$1.25 \left(1 + \frac{R_2}{R_1} \right)$

See [Figure 11](#)

Title

9.2.3.3 Application Performance

Table

TEST	CONDITIONS	RESULTS
Line regulation	$V_{IN} = 15 \text{ V to } 25 \text{ V}, I_O = 500 \text{ mA}$	$12 \text{ mV} \pm 0.12\%$
Load regulation	$V_{IN} = 25 \text{ V}, I_O = 50 \text{ mA to } 500 \text{ mA}$	$3 \text{ mV} \pm 0.03\%$
Output ripple	$V_{IN} = 25 \text{ V}, I_O = 500 \text{ mA}$	120 mV_{pp}
Short-circuit current	$V_{IN} = 25 \text{ V}, R_L = 0.1 \Omega$	1.1 A
Efficiency	$V_{IN} = 25 \text{ V}, I_O = 500 \text{ mA}$	83.7%
Output ripple with optional filter	$V_{IN} = 25 \text{ V}, I_O = 500 \text{ mA}$	40 mV_{pp}

Page

Narrative Text

Narrative Text

10 Power Supply Recommendations

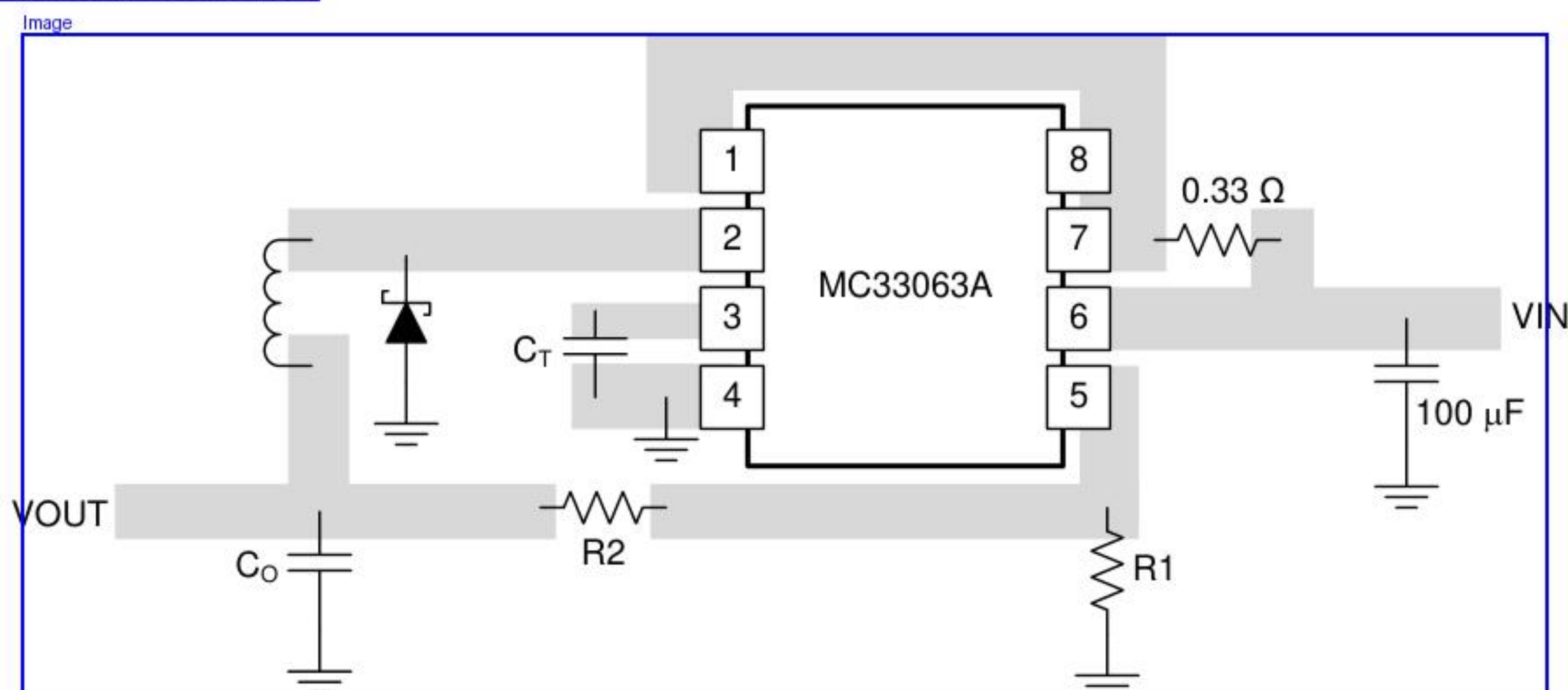
Narrative Text
 This device accepts 3 V to 40 V on the input. It is recommended to have a 1000- μ F decoupling capacitor on the input.

11 Layout

11.1 Layout Guidelines

Narrative Text
 Keep feedback loop layout trace lengths to a minimum to avoid unnecessary IR drop. In addition, the loop for the decoupling capacitor at the input should be as small as possible. The trace from V_{IN} to pin 1 of the device should be thicker to handle the higher current.

11.2 Layout Example



FigureCaption
Figure 12. Layout Example for a Step-Down Converter

Header

MC33063A, MC34063A

Narrative Text

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12 Device and Documentation Support

Title

12.1 Related Links

Narrative Text

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

FigureCaption

Table 1. Related Links

Table

PARTS	PRODUCT FOLDER	SAMPLE & BUY	TECHNICAL DOCUMENTS	TOOLS & SOFTWARE	SUPPORT & COMMUNITY
MC33063A	Click here	Click here	Click here	Click here	Click here
MC34063A	Click here	Click here	Click here	Click here	Click here

Title

12.2 Trademarks

Narrative Text

All trademarks are the property of their respective owners.

Title

12.3 Electrostatic Discharge Caution

Uncategorized Text

Narrative Text



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

Title

12.4 Glossary

Narrative Text

[SLYZ022](#) — TI Glossary.

Narrative Text

This glossary lists and explains terms, acronyms, and definitions.

Title

13 Mechanical, Packaging, and Orderable Information

Narrative Text

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

LinkItem

18 [Submit Documentation Feedback](#)

Narrative Text

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Narrative Text

Product Folder Links: [MC33063A](#) [MC34063A](#)

Title
PACKAGING INFORMATION

Table

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
MC33063AD	Last Time Buy	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	M33063A
MC33063AD.A	Last Time Buy	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	M33063A
MC33063ADE4	Last Time Buy	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	M33063A
MC33063ADG4	Last Time Buy	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	M33063A
MC33063ADR	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	M33063A
MC33063ADR.A	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	M33063A
MC33063ADRE4	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	M33063A
MC33063ADRG4	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	M33063A
MC33063ADRJR	Last Time Buy	Production	SON (DRJ) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 85	ZYF
MC33063ADRJR.A	Last Time Buy	Production	SON (DRJ) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 85	ZYF
MC33063AP	Last Time Buy	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	MC33063AP
MC33063AP.A	Last Time Buy	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	MC33063AP
MC33063APE4	Last Time Buy	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	MC33063AP
MC34063AD	Last Time Buy	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	M34063A
MC34063AD.A	Last Time Buy	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	M34063A
MC34063ADE4	Last Time Buy	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	M34063A
MC34063ADG4	Last Time Buy	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	M34063A
MC34063ADR	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	M34063A
MC34063ADR.A	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	M34063A

Table

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
MC34063ADRJR	Last Time Buy	Production	SON (DRJ) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	0 to 70	ZYF
MC34063ADRJR.A	Last Time Buy	Production	SON (DRJ) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	0 to 70	ZYF
MC34063ADRJRG4	Last Time Buy	Production	SON (DRJ) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	0 to 70	ZYF
MC34063AP	Last Time Buy	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	MC34063AP
MC34063AP.A	Last Time Buy	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	MC34063AP
MC34063APE4	Last Time Buy	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	MC34063AP

Narrative Text

(1) **Status:** For more details on status, see our [product life cycle](#).

Narrative Text

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

Title

(3) **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

Narrative Text

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

Narrative Text

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

Narrative Text

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Narrative Text

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

Narrative Text

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Narrative Text

and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

Narrative Text

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

Title

OTHER QUALIFIED VERSIONS OF MC33063A :

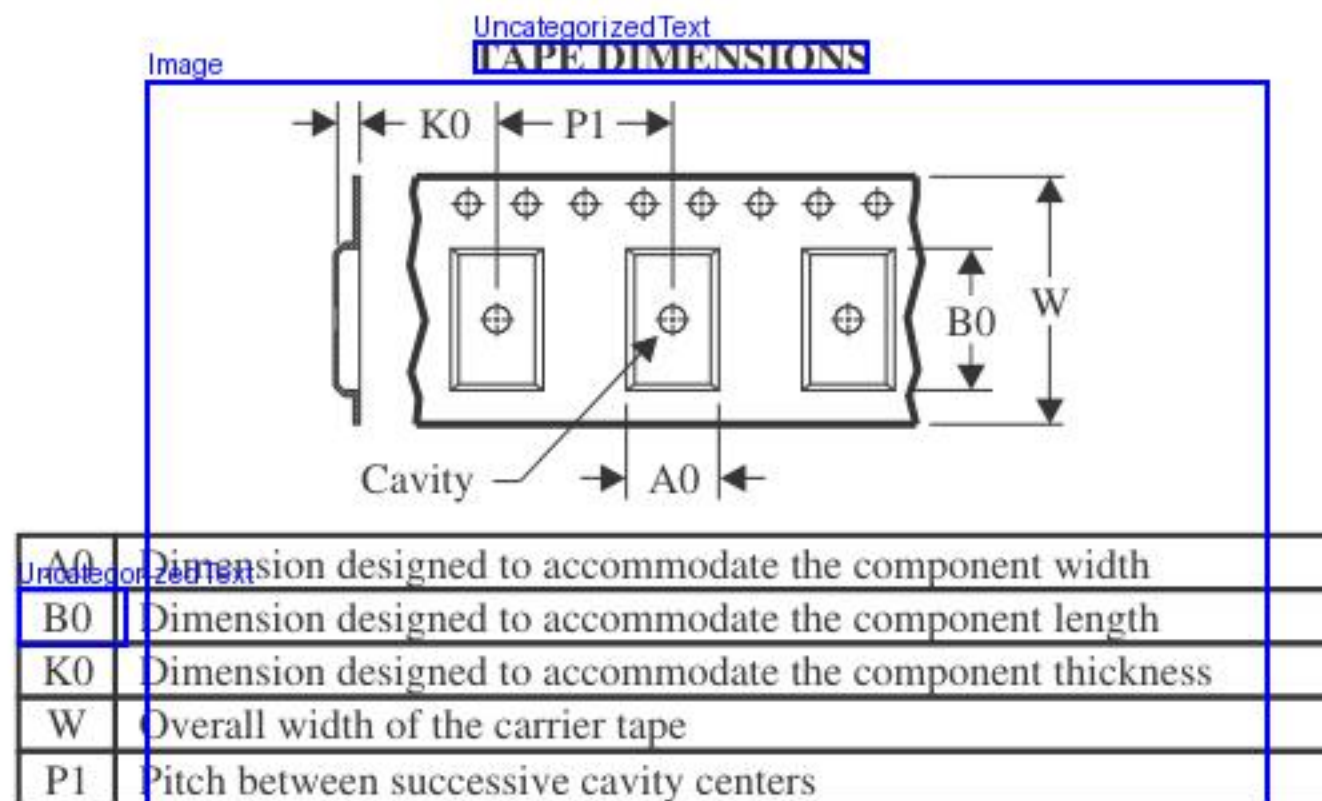
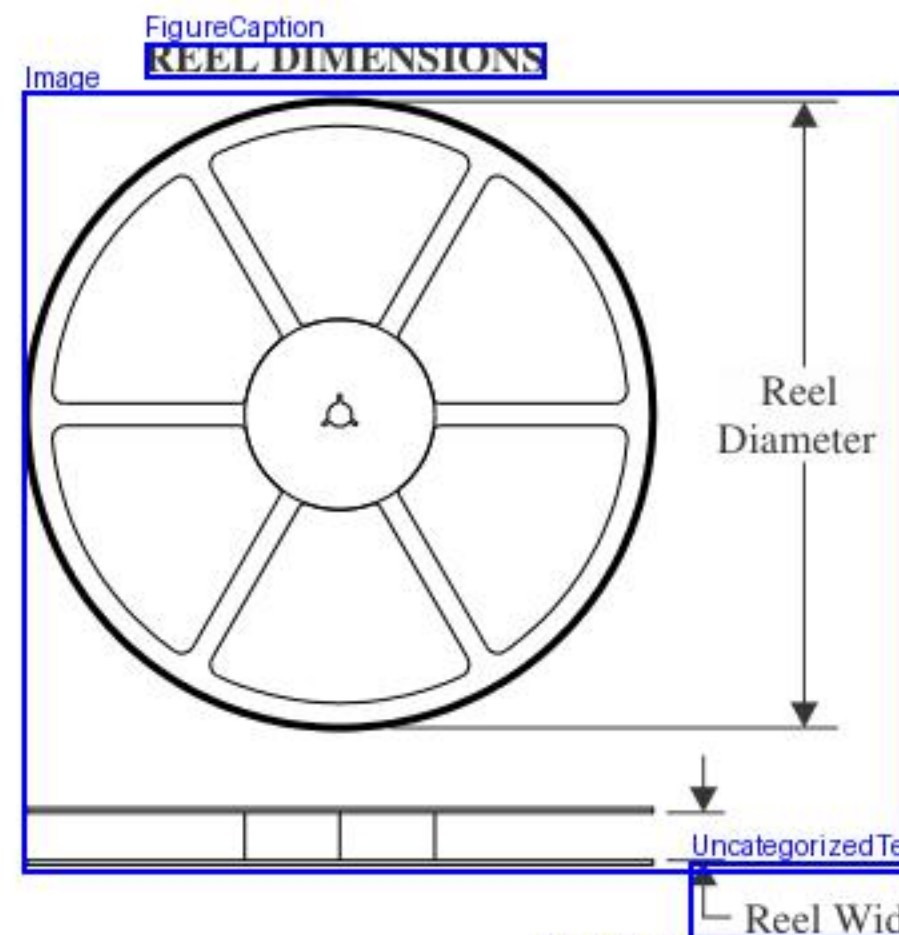
- Narrative Text
- Automotive : MC33063A-Q1

Narrative Text

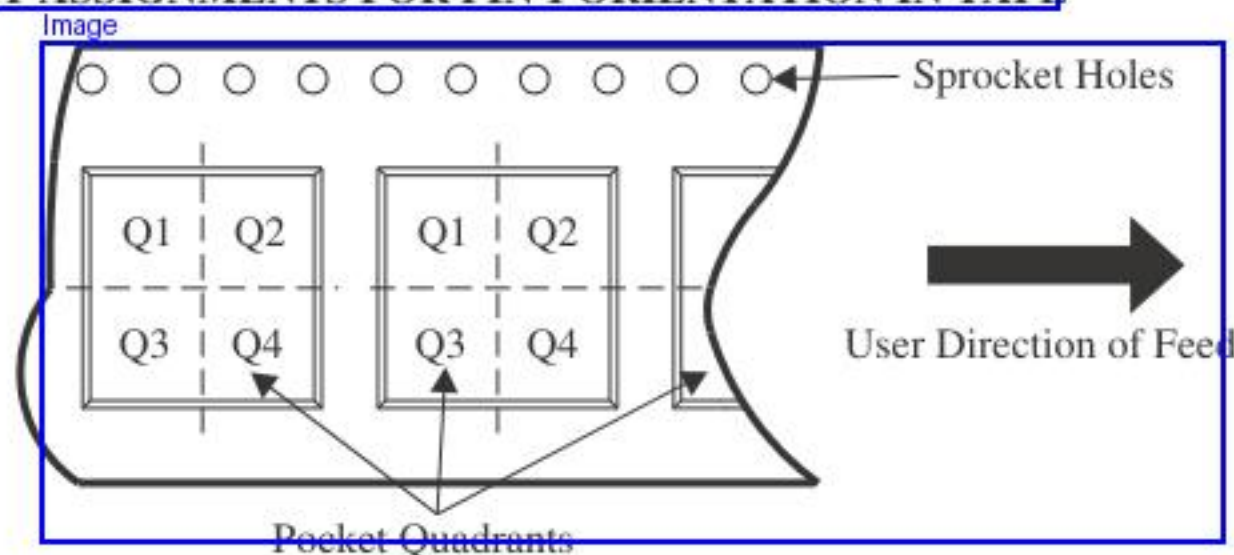
NOTE: Qualified Version Definitions:

- Narrative Text
- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

Title
TAPE AND REEL INFORMATION



NarrativeText
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE

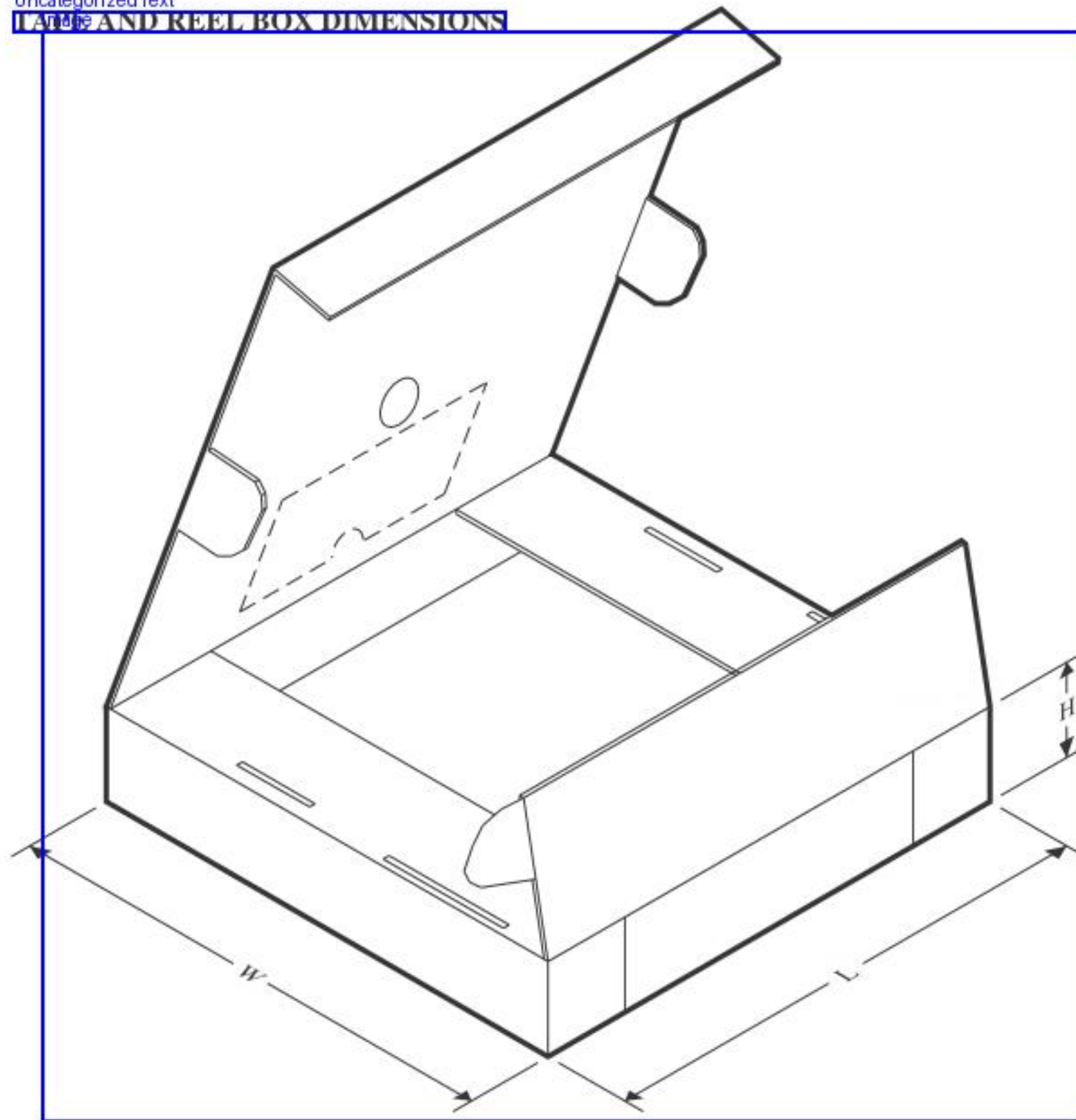


FigureCaption
 All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
MC33063ADR	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
MC33063ADR	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
MC33063ADRJR	SON	DRJ	8	1000	330.0	12.4	4.25	4.25	1.15	8.0	12.0	Q2
MC34063ADR	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1

UncategorizedText

TAPE AND REEL BOX DIMENSIONS

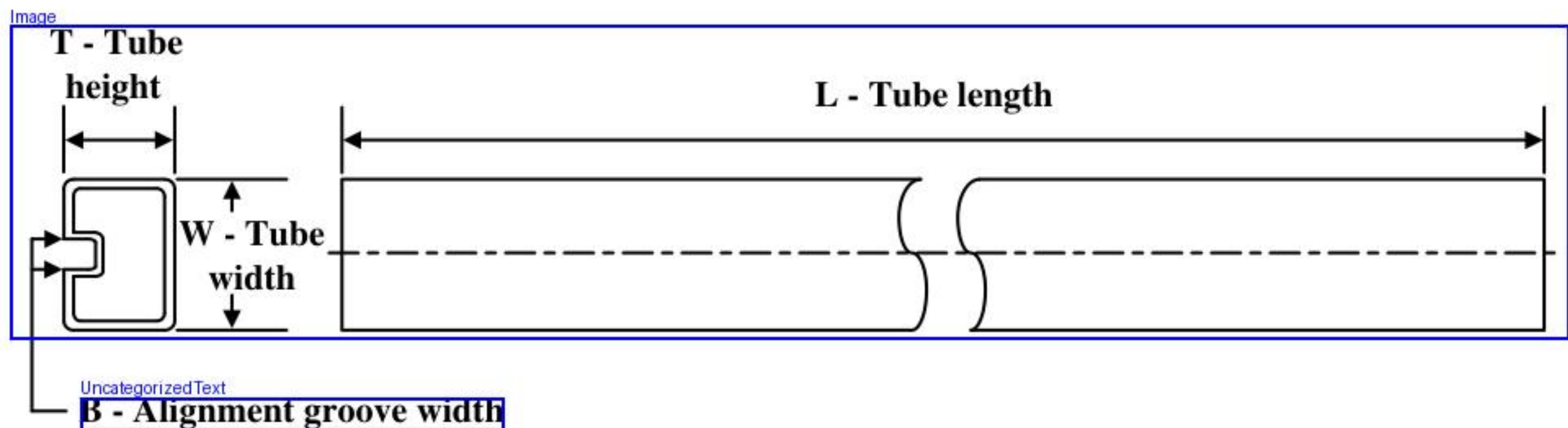


FigureCaption

All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
MC33063ADR	SOIC	D	8	2500	340.5	338.1	20.6
MC33063ADR	SOIC	D	8	2500	353.0	353.0	32.0
MC33063ADRJR	SON	DRJ	8	1000	353.0	353.0	32.0
MC34063ADR	SOIC	D	8	2500	353.0	353.0	32.0

Title
TUBE



UncategorizedText
 All dimensions are nominal

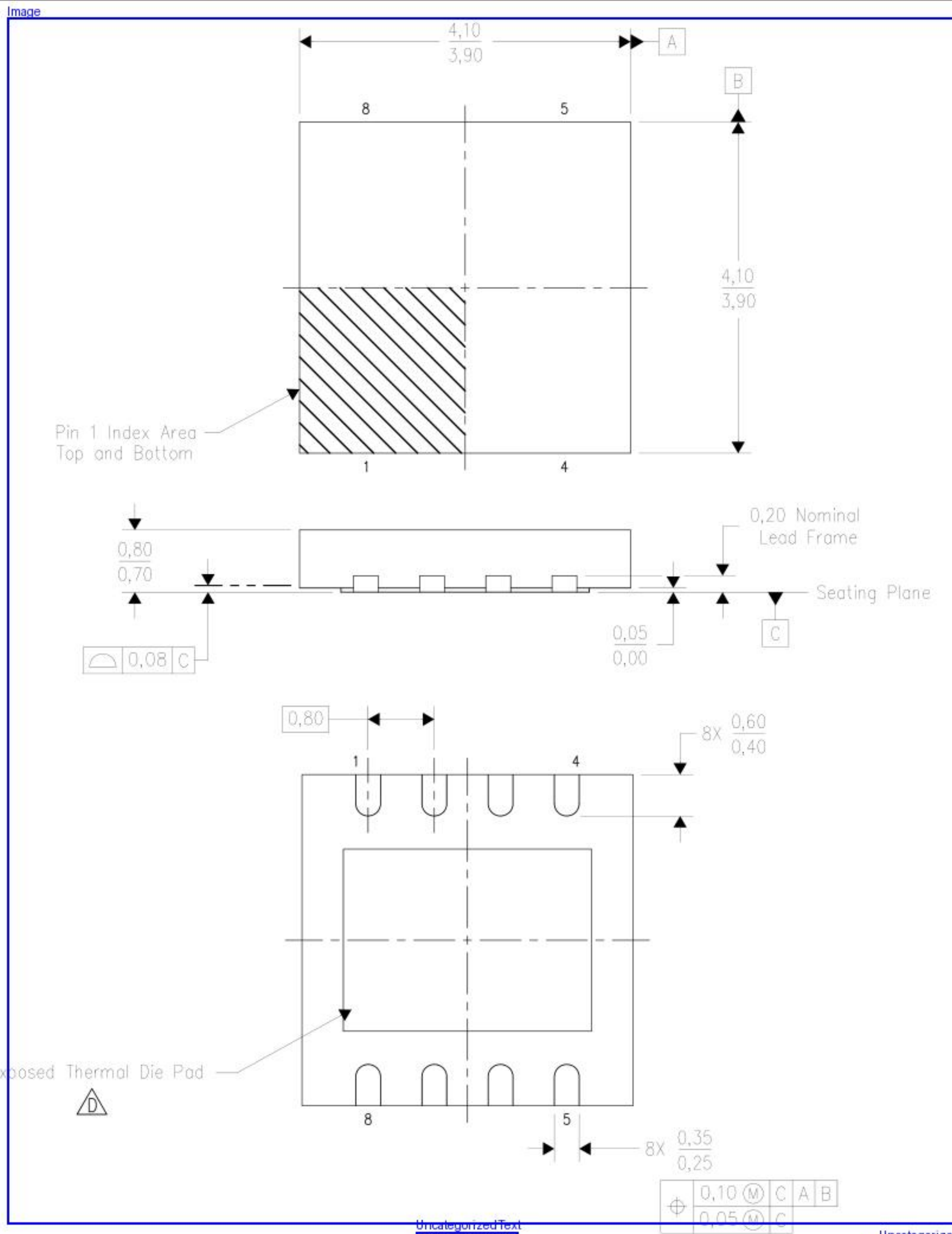
Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
MC33063AD	D	SOIC	8	75	507	8	3940	4.32
MC33063AD	D	SOIC	8	75	506.6	8	3940	4.32
MC33063AD.A	D	SOIC	8	75	507	8	3940	4.32
MC33063AD.A	D	SOIC	8	75	506.6	8	3940	4.32
MC33063ADE4	D	SOIC	8	75	507	8	3940	4.32
MC33063ADE4	D	SOIC	8	75	506.6	8	3940	4.32
MC33063ADG4	D	SOIC	8	75	507	8	3940	4.32
MC33063ADG4	D	SOIC	8	75	506.6	8	3940	4.32
MC33063AP	P	PDIP	8	50	506	13.97	11230	4.32
MC33063AP.A	P	PDIP	8	50	506	13.97	11230	4.32
MC33063APE4	P	PDIP	8	50	506	13.97	11230	4.32
MC34063AD	D	SOIC	8	75	507	8	3940	4.32
MC34063AD.A	D	SOIC	8	75	507	8	3940	4.32
MC34063ADE4	D	SOIC	8	75	507	8	3940	4.32
MC34063ADG4	D	SOIC	8	75	507	8	3940	4.32
MC34063AP	P	PDIP	8	50	506	13.97	11230	4.32
MC34063AP.A	P	PDIP	8	50	506	13.97	11230	4.32
MC34063APE4	P	PDIP	8	50	506	13.97	11230	4.32

UncategorizedText

DRJ (S-PWSON-N8)

UncategorizedText

PLASTIC SMALL OUTLINE NO-LEAD

UncategorizedText
Bottom View

UncategorizedText

#205439/C 12/10

NOTES:

A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M-1994.

B. This drawing is subject to change without notice.

C. SON (Small Outline No-Lead) package configuration.

D. The package thermal pad must be soldered to the board for thermal and mechanical performance.

E. See the Product Data Sheet for details regarding the exposed thermal pad dimensions.

F. Package complies to JEDEC MO-229 variation WGGB.

Image

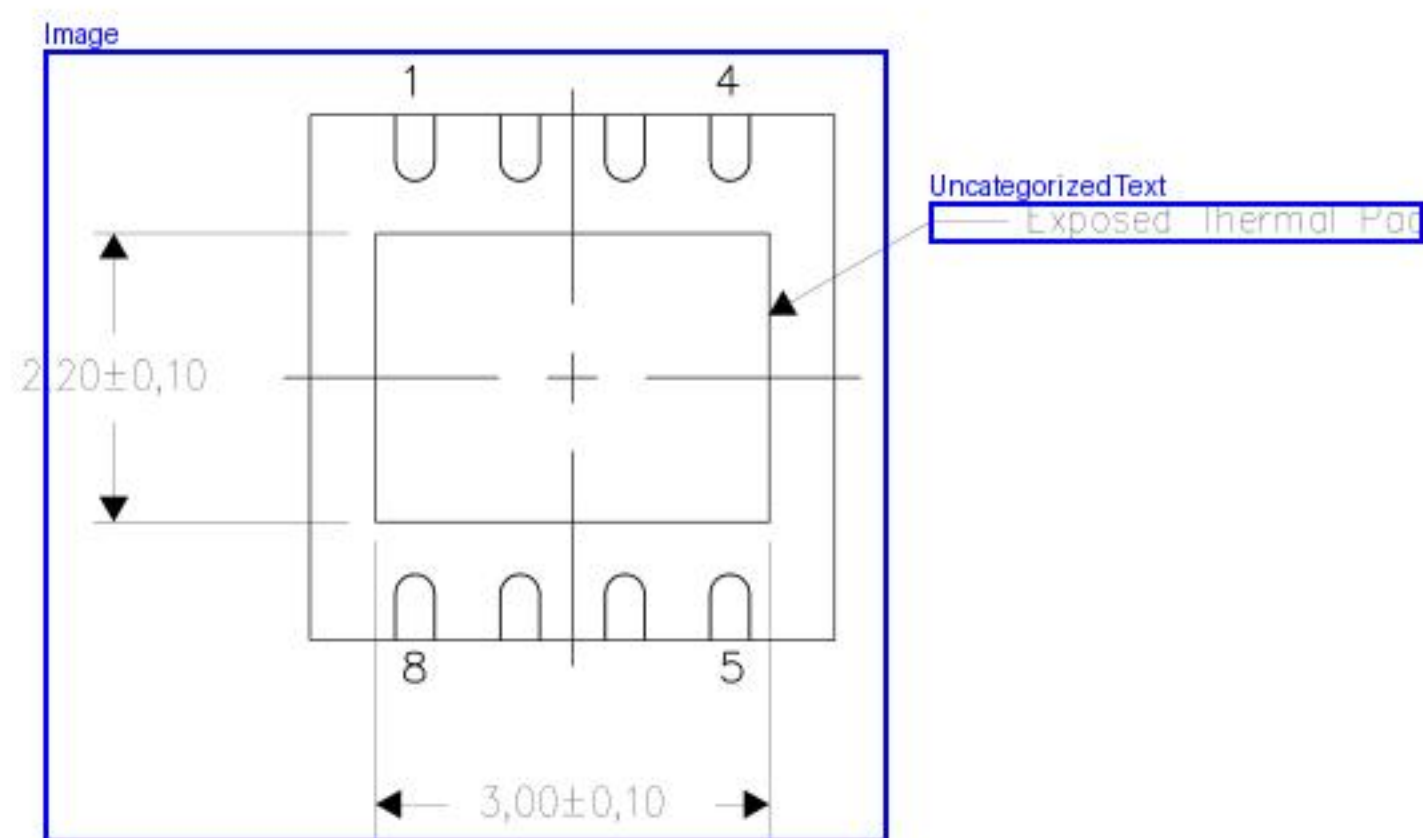
DRJ (S-PWSON-N8)	PLASTIC SMALL OUTLINE NO-LEAD
------------------	-------------------------------

Title
THERMAL INFORMATION

Narrative Text
This package incorporates an exposed thermal pad that is designed to be attached directly to an external heatsink. The thermal pad must be soldered directly to the printed circuit board (PCB). After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

Narrative Text
For information on the Quad Flatpack No-Lead (QFN) package and its advantages, refer to Application Report, QFN/SON PCB Attachment, Texas Instruments Literature No. SLUA271. This document is available at www.ti.com.

Narrative Text
The exposed thermal pad dimensions for this package are shown in the following illustration.



FigureCaption
Bottom View

FigureCaption
Exposed Thermal Pad Dimensions

NarrativeText
4206882/F 01/11

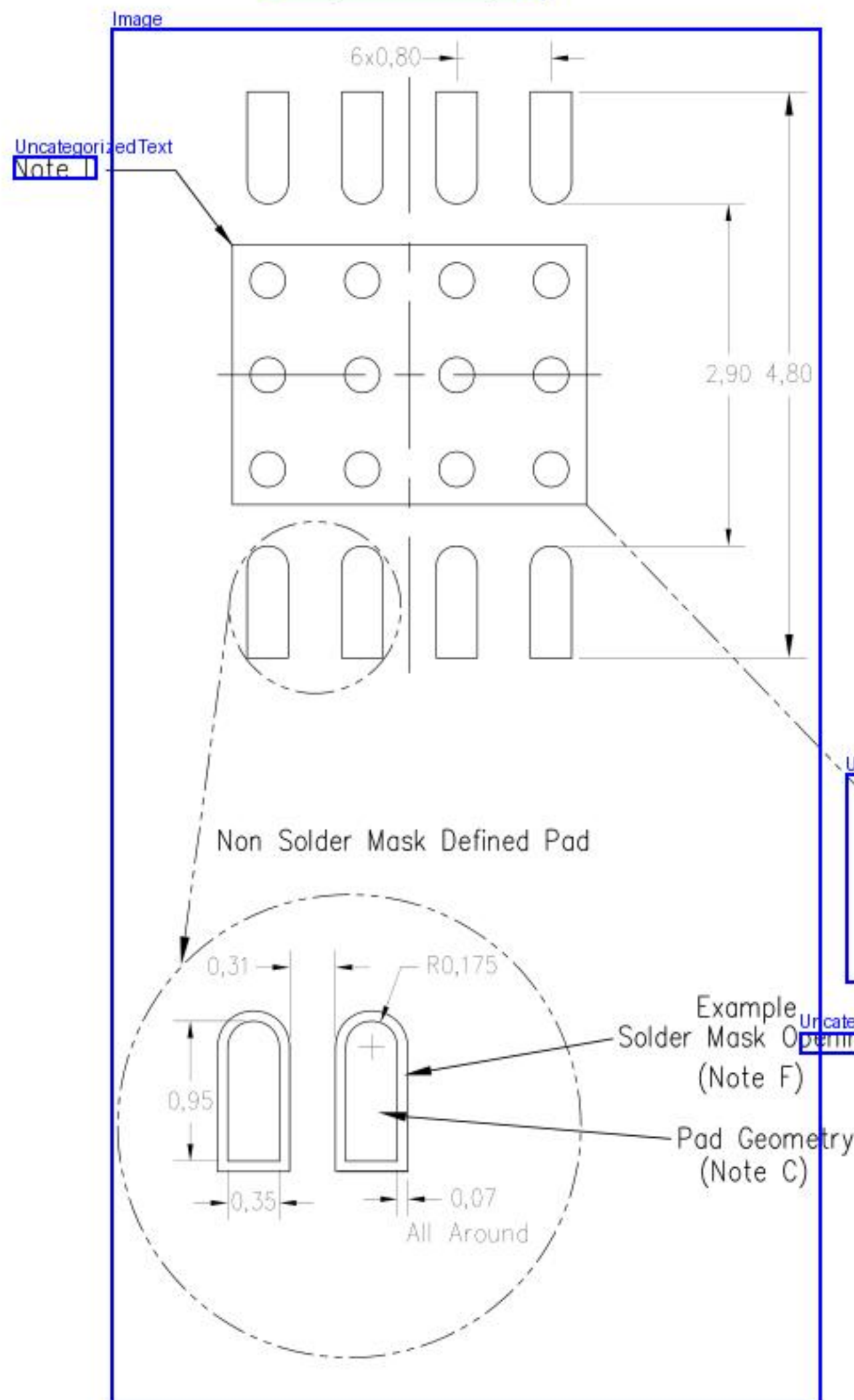
ListItem
NOTE: All linear dimensions are in millimeters

UncategorizedText

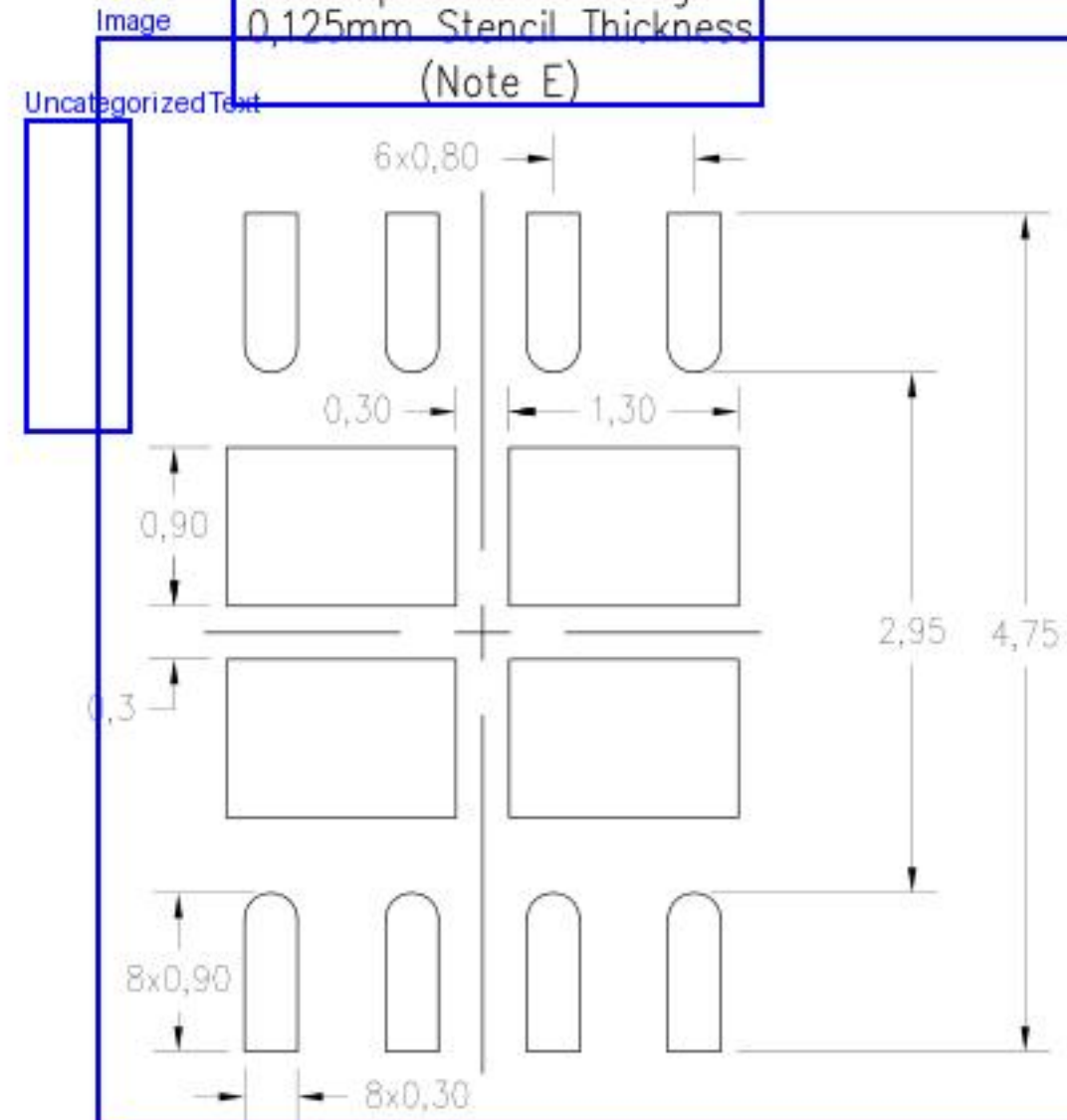
DRJ (S-PWSON-N8)

UncategorizedText

SMALL PACKAGE OUTLINE NO-LEAD

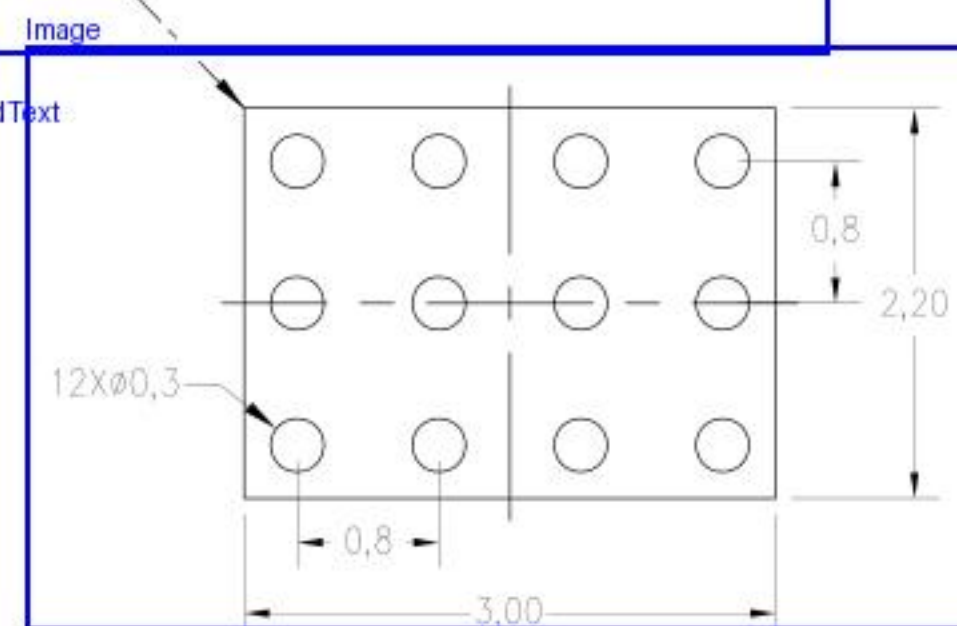
FigureCaption
Example Board Layout

NarrativeText

Example Stencil Design
0,125mm Stencil Thickness
(Note E)FigureCaption
71% Printed Solder Coverage by Area

UncategorizedText

Example Via Layout Design
may vary depending on constraints
(Note D, F)



UncategorizedText

4208469/D 01/1

ListItem

NOTES: A. All linear dimensions are in millimeters.

B. This drawing is subject to change without notice.

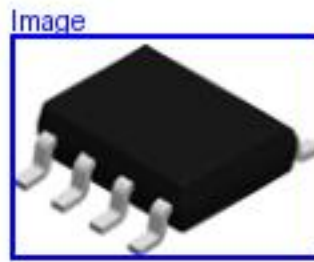
C. Publication IPC-7351 is recommended for alternate designs.

D. This package is designed to be soldered to a thermal pad on the board. Refer to Application Note, Quad Flat-Pack Packages, Texas Instruments Literature No. SLUA271, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <<http://www.ti.com>>.

E. Laser cutting apertures with electropolish and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC 7525 for stencil design considerations.

F. Customers should contact their board fabrication site for solder mask tolerances and via tenting recommendations for vias placed in the thermal pad.

Image

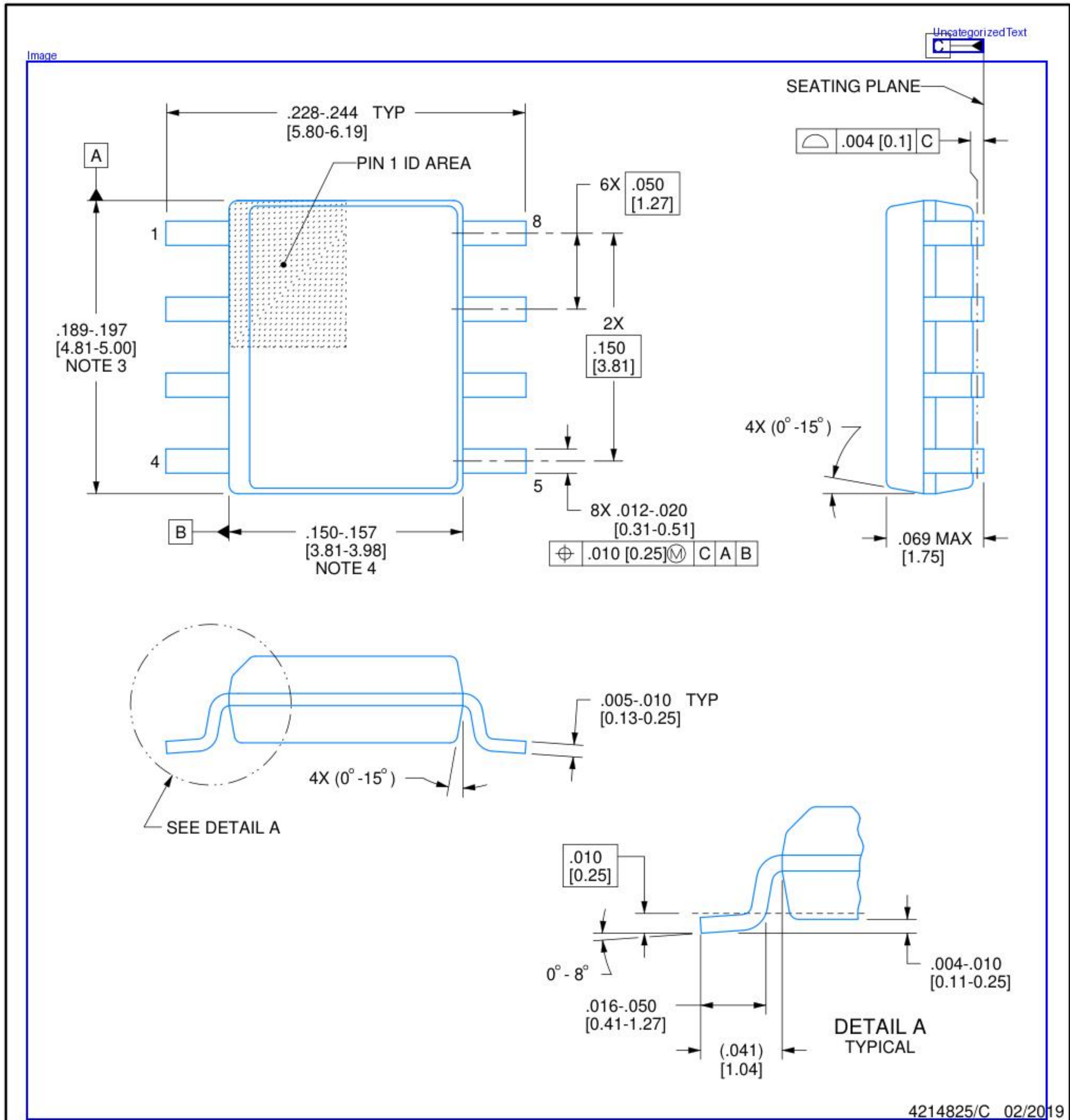


PACKAGE OUTLINE

Title
D0008A

Title
SOIC - 1.75 mm max height

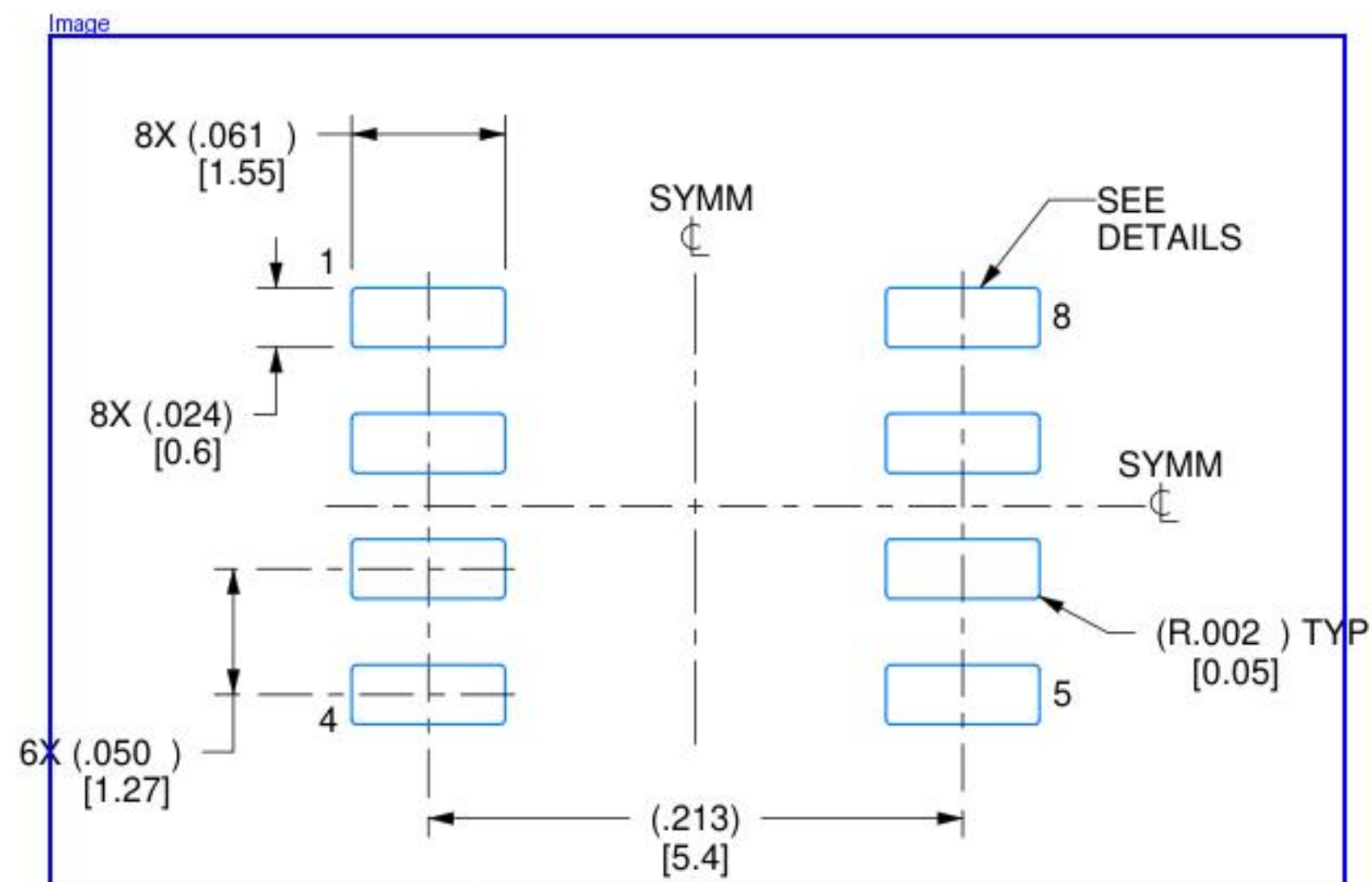
Narrative Text
SMALL OUTLINE INTEGRATED CIRCUIT



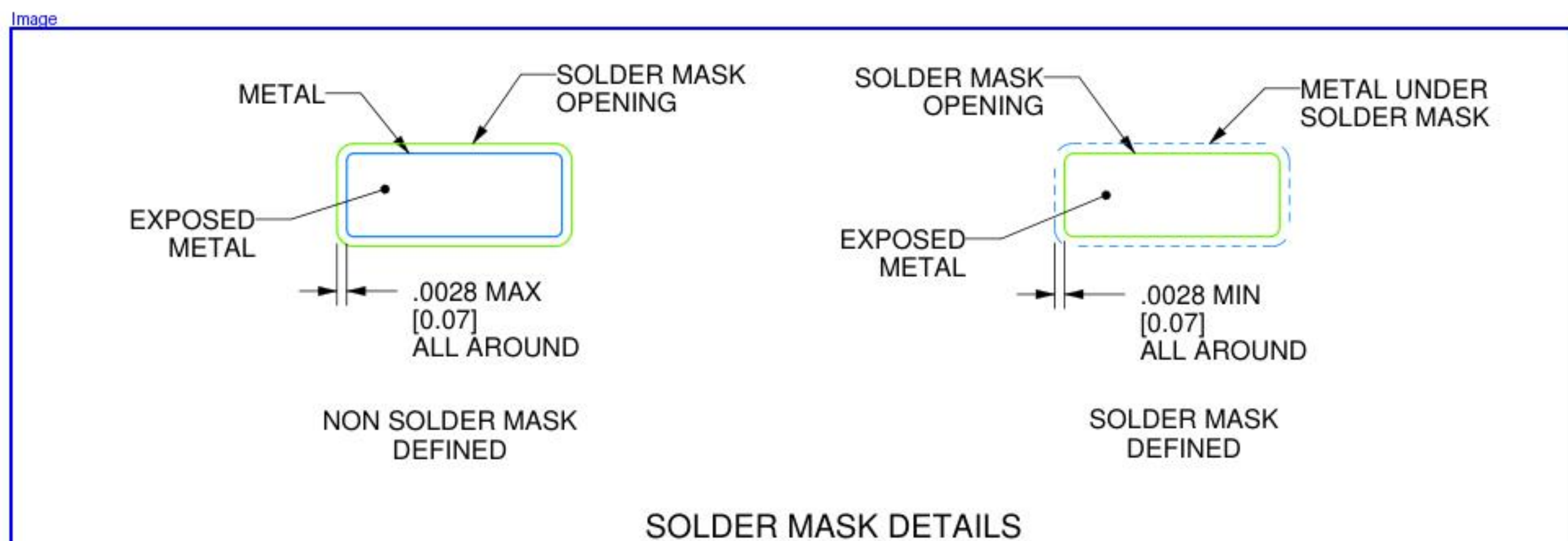
4214825/C 02/2019

NOTES:

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.



FigureCaption
LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X



4214825/C 02/2019

Narrative Text
NOTES: (continued)

List Item
6. Publication IPC-7351 may have alternate designs.

List Item
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

Title

EXAMPLE STENCIL DESIGN

Header

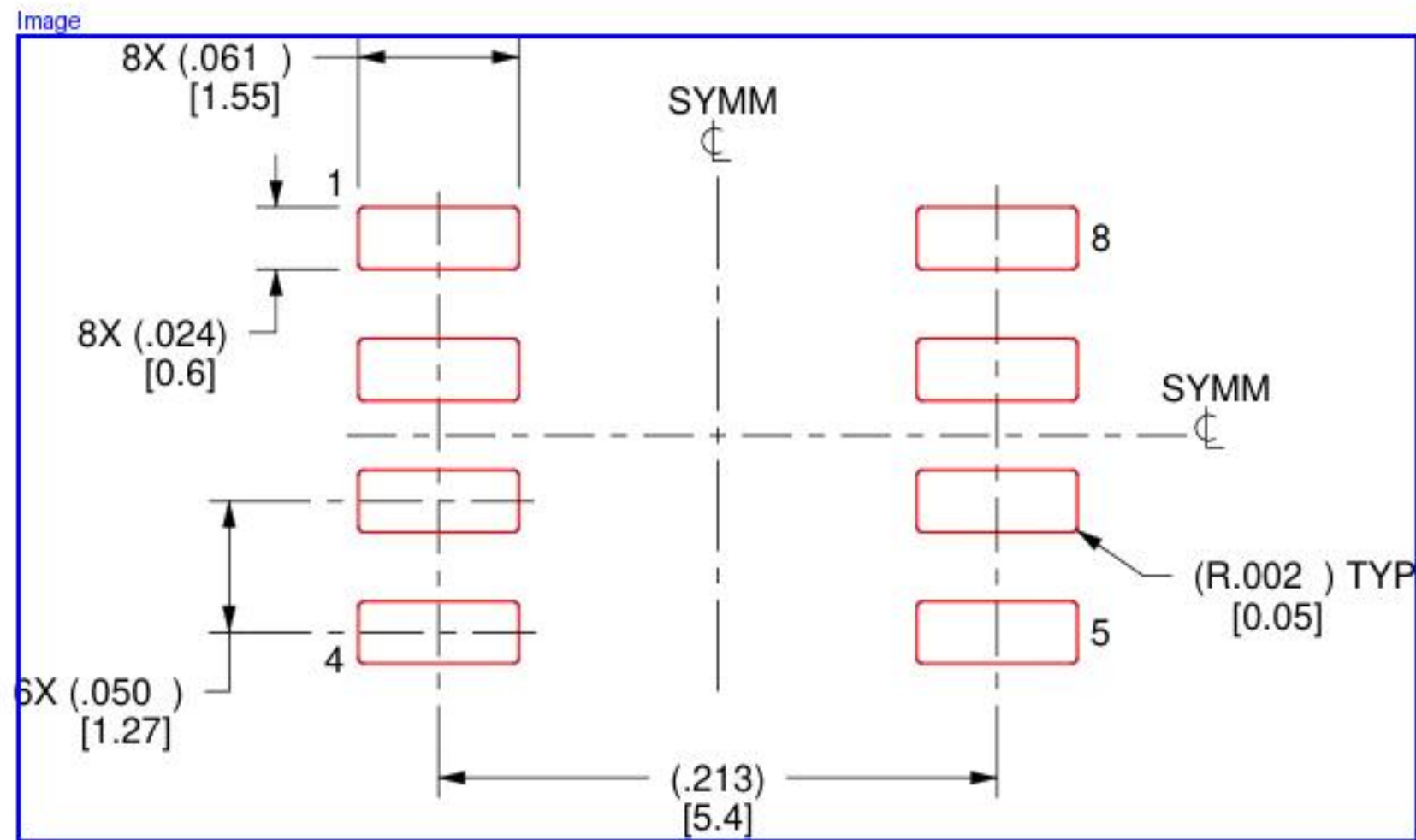
D0008A

Title

SOIC - 1.75 mm max height

Narrative Text

SMALL OUTLINE INTEGRATED CIRCUIT



FigureCaption

SOLDER PASTE EXAMPLE
 BASED ON .005 INCH [0.125 MM] THICK STENCIL
 SCALE:8X

Header

4214825/C 02/2019

Narrative Text

NOTES: (continued)

ListItem

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

9. Board assembly site may have different recommendations for stencil design.

Image

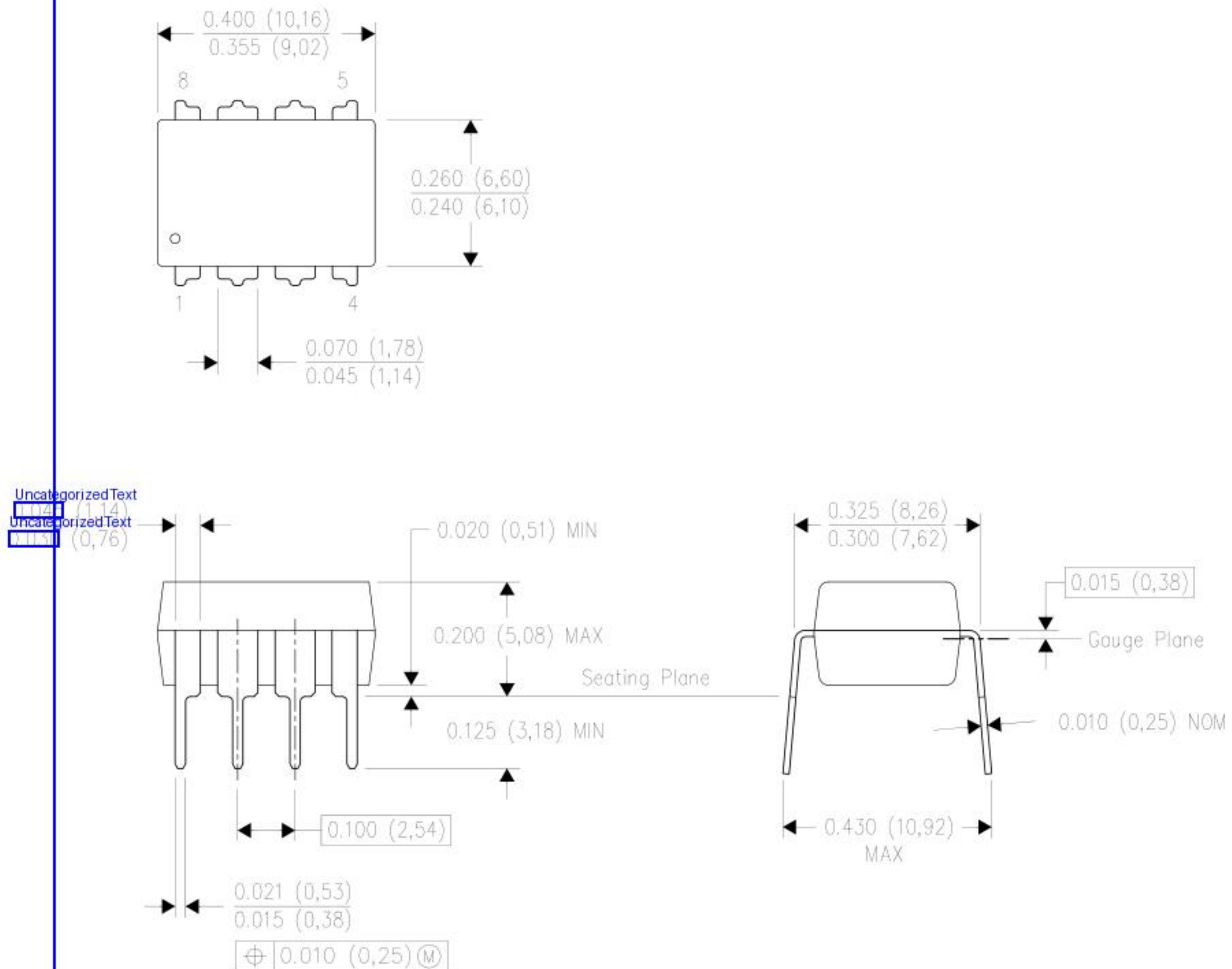
UncategorizedText

P (R-PDIP-T8)

Narrative Text

PLASTIC DUAL-IN-LINE PACKAGE

Image



UncategorizedText

4040082/E 04/2010

UncategorizedText

NOTES

- A. All linear dimensions are in inches (millimeters).
- B. This drawing is subject to change without notice.
- C. Falls within JEDEC MS-001 variation BA.

Image

Title

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